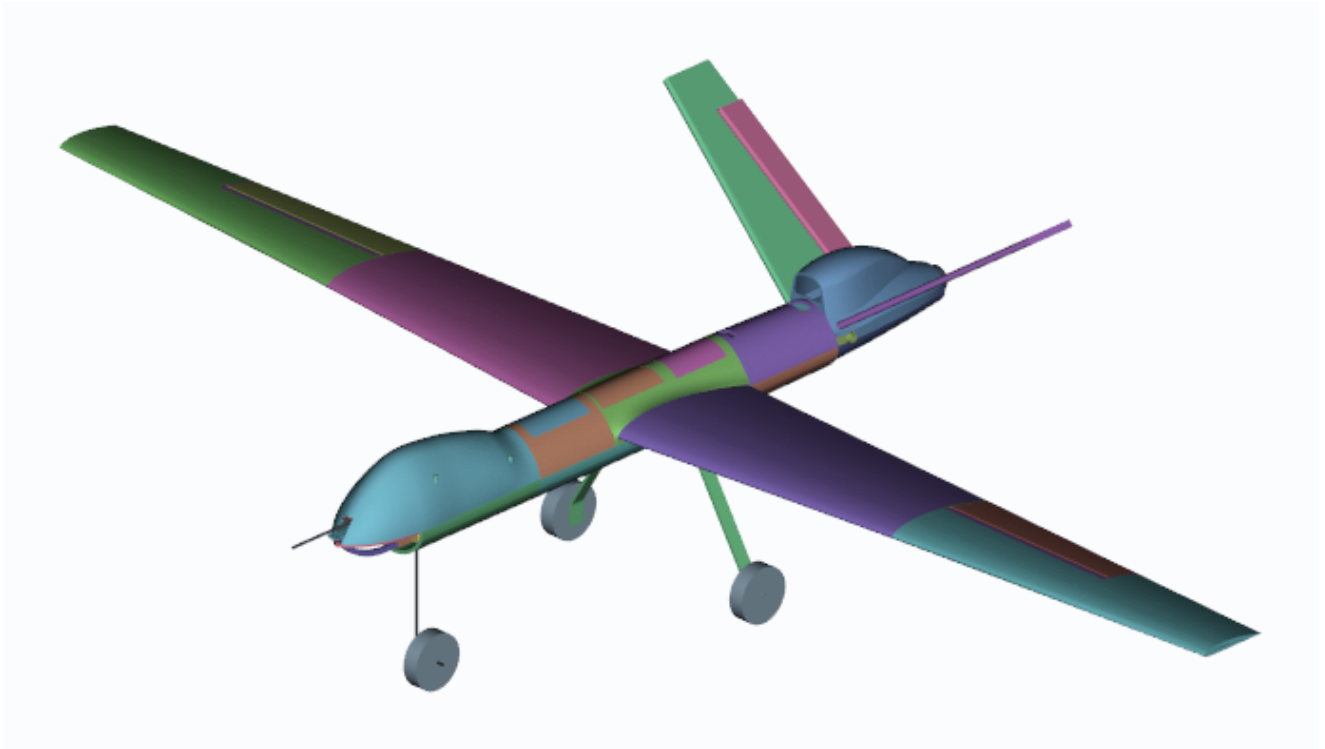


MQ9 / complete build guide

Manufacturing Team + Electronics Team / Revision 2



Use the contents or PDF bookmarks to jump to a task. Read the weight/flight assessment, then the body and electronics chapters. Reference tables and atlases follow. Team PDFs repeat these chapters.

Actual-size cutting templates remain in Manufacturing /Print_Templates. Do not cut parts from the reduced illustrations in this manual.

Contents

Page	Chapter
3	Weight and flight assessment
8	Manufacturing Team
23	Electronics Team
40	Parts and materials
45	Hardware and forming
46	Foam panel map
51	Foam forming
54	Plywood profile atlas
59	Foam pattern overview
65	Plywood sheet overview
68	Foam sheet layouts
71	Motor mount pattern
72	Checks and acceptance

Before the full batch

Run the process coupons, dry-fit actual hardware and agree access before closing the body. Record masses and physical checks. File checks do not establish flight readiness.

Weight and printing

Paper-review result: cruise is plausible under the stated assumptions. Flight and hand launch remain on hold until stability/trim, propulsion and structural checks close. This is a screening calculation, not a flight clearance.

Installed category	Nominal mass
Printed parts	728 g
Plywood	243 g
Foam	337 g
Metal landing gear	144 g
Battery	512 g
Other equipment, tubes, wiring, glue and hardware	669 g
Total	2.63 kg

The estimate includes a 25 g propeller/nut allowance, 90 g glue/tape, 90 g fasteners/linkages, 80 g wiring, 70 g tubes/retainers and 75 g wheels. These are allowances to replace with actual weights. Plywood assumes 0.55 g/cm³; unpeeled foam assumes 400 g/m².

What is printed

Group	Installed plastic
Nose shell A59	150 g
Centre fairing A06 / tail housing A12	65 g / 122 g
Two ailerons / two hinge rails	115 g / 56 g
Two wing-root fittings / nose gear mount and bearing	56 g / 19 g
Mounts, spacers, washers, hatches and supports	144 g

There are 38 aircraft STL files for 42 installed pieces. The fit coupon is additional. Object extrusion totals 728 g; filament with automatic supports is 1.30 kg, or 1.50 kg with 15% waste allowance. Budget two 1 kg spools.

PrusaSlicer estimate: 0.4 mm nozzle, 0.2 mm layers, 4 top/bottom layers; shells 2 walls/5%, controls 3 walls/10%, carriers 4 walls/30%, automatic supports. A12 uses the shell setting only in this estimate; its structural process is not qualified. Reorient/reslice for the actual printer and reweigh. PLA+ is not the LW-PLA used in the video.

Balance, lift and stability

Wing area 0.30445 m²; span 2.000 m; mean aerodynamic chord 169.83 mm; aspect ratio 13.14. At 2.633 kg, weight is 25.83 N and wing loading is 86.5 g/dm².

Battery centre Y	Calculated CG	Fraction of MAC
-60 mm / forward stop	517.2 mm aft nose	22.0%
-33.7 mm	522.3 mm aft nose	25.0%
+20 mm / aft stop	532.8 mm aft nose	31.2%

CG = sum(mass x station) / total mass. The table is adjustment capability, **not an approved CG range**. Only about 36.6 g added at Y450 would consume the remaining ability to reach geometric quarter-MAC with the battery fully forward. Finish weighing before fixing the battery stops.

Assumed maximum lift coefficient	Stall speed	1.3 x stall reference
0.8	13.16 m/s	17.10 m/s
1.0	11.77 m/s	15.30 m/s
1.2	10.74 m/s	13.97 m/s

Stall speed = $\sqrt{2 \times \text{weight} / (\text{air density} \times \text{wing area} \times \text{maximum lift coefficient})}$. Density is 1.225 kg/m³ here. At 1.0 kg/m³, speeds increase by 10.7%. The lift coefficients are assumed; low-Reynolds-number airfoil behaviour, paper texture, twist and forming accuracy are not measured. Reynolds number is about 135,000 at 12 m/s and 230,000 at 20 m/s.

Tail and stability screen

The V-tail has about 0.0501 m² effective pitch area, 0.0246 m² effective yaw area, 35-degree cant and a 0.368 m tail arm. Pitch tail-volume coefficient is 0.357; vertical volume is 0.0149.

The linear model is SM = tail efficiency x lift-slope ratio x (1 - downwash gradient) x tail volume - body allowance - CG offset / MAC. Assumed factors are 0.8-1.0, 0.65-0.9, 0.25-0.5 and 0.03-0.10 MAC respectively.

Battery position	Static-margin sensitivity
Forward stop	+2.3% to +24.1% MAC
Geometric quarter-MAC	-0.7% to +21.1% MAC
Aft stop	-6.9% to +15.0% MAC

These are extreme assumption cases, not measured stability limits. They neither establish a safe aft limit nor prove the actual aircraft unstable. Resolve neutral point, trim, incidence and combined V-tail authority before flight.

Power, launch and flight time

Selected hardware remains Emax ECO II 2807 1300KV, Gemfan 9045CL three-blade, 4S 5200 mAh and a 40 A ESC. No verified thrust/current map for this exact combination is available in this package. The selection is a build assumption; it has not passed a propulsion test.

Nominal cruise scenario	Electrical power	Current at 14 V
16 m/s	61-147 W	4.4-10.5 A
20 m/s	98-257 W	7.0-18.3 A

$q = 0.5 \times \text{density} \times \text{speed squared}$; $CL = \text{weight} / (q \times \text{area})$; $CD = CD0 + CL \text{ squared} / (\pi \times \text{efficiency factor} \times \text{aspect ratio})$; $\text{drag} = q \times \text{area} \times CD$; $\text{electrical power} = \text{drag} \times \text{speed} / \text{propulsion efficiency} + 10 \text{ W avionics}$.

Assumptions: zero-lift drag coefficient 0.035-0.070, span efficiency 0.65-0.85 and combined propulsion efficiency 0.45-0.65. These broad scenarios are not a current, climb-rate or thrust prediction. Loaded motor RPM and propeller thrust at flying speed remain unknown.

Launch airspeed	Lift coefficient needed for 1g flight
8 m/s	2.16
10 m/s	1.39
12 m/s	0.96
14 m/s	0.71

An 8-10 m/s throw is not sufficient for steady 1g flight if maximum lift is within the assumed 0.8-1.2 range. Acceleration and height loss depend on thrust versus airspeed, attitude, wind and technique. An autopilot launch trigger is not flying speed. No ordinary hand launch is cleared by this calculation.

Flight-time planning

Average total battery current	Time with 20% nominal charge reserve
8 A / 12 A	31.2 min / 20.8 min
16 A / 20 A	15.6 min / 12.5 min
25 A / 40 A	10.0 min / 6.2 min

Planning range: about 12-20 minutes only if average total current is 12-20 A. $\text{Time} = 5.2 \text{ Ah} \times 0.8 \times 60 / \text{current} = 249.6 / \text{current}$, in minutes. Nominal battery energy is 76.96 Wh; 80% is 61.57 Wh. Capacity loss, climb, wind and return reserve affect the result. Use calibrated flight logs to select a real endurance estimate.

Controls and structural loads

MG90S manufacturer stall torque at 4.8 V is 1.8 kgf-cm, or 0.1765 Nm. The screening allowance below is 50% of stall: 0.0883 Nm. This is an assumption, not a continuous servo rating.

Control at 20 m/s	Hinge torque	Servo torque
Aileron: 0.01787 m ² , 74.3 mm chord	0.0651 Nm	0.0434 Nm
Moving tail: 0.01060 m ² , 40 mm chord	0.0208 Nm	0.0139 Nm

Hinge moment = $q \times \text{control area} \times \text{control chord} \times \text{hinge coefficient}$. Assumed hinge coefficient is 0.2; servo arm is 10 mm and control horn 15 mm. Servo torque = hinge moment $\times 10/15$. Aileron demand rises to 0.0678 Nm at 25 m/s and 0.0976 Nm at 30 m/s, above the screening allowance. No speed or flutter limit is established.

An ideal straight linkage gives about 28 degrees of control rotation at 45 degrees servo rotation, or 35 degrees at 60 degrees. Real horn alignment, friction and combined V-tail mixing change this. Verify simultaneous pitch and yaw commands, mechanical stops and voltage at the farthest servo. Vx is rated 5 A continuous/6 A peak; the separate peripheral 5 V rail is 2 A.

Wing and joiners

Screening case	Demand
Uniform 1g lift / each half-wing root	6.46 Nm
Uniform 4g lift / each half-wing root	25.83 Nm
Elliptical 1g / 4g lift	5.48 / 21.92 Nm
Two 8/6 mm tubes sharing 4g moment equally	About 376 MPa per tube
One tube carrying that entire moment	About 752 MPa

Uniform root moment = total weight $\times$ half-span / 4. A single 8 mm OD / 6 mm ID tube has elastic section modulus 34.36 mm³; stress = moment / section modulus. These are load illustrations, not the actual load split or tube strength. Spars, sleeves, bonds, printed roots and skins also carry load. Tube layup, joint stiffness and crush strength require evidence; 5 mm solid rods are not substitutes.

A12 is a structural tail housing

The fixed tail meets A12 and is about 3 mm from plywood bulkhead A18. Tail loads therefore need a verified transfer through the printed slot walls and bonded housing into the frame. A12 also supports two servo mounts. Do not treat its 2-wall/5% slicing estimate as a qualified production recipe. Check local wall orientation, root bonds, fasteners and the A12-to-A18 joint using representative coupons and the assembled tail.

Landing loads and closure checks

Current 76.2 mm wheel gear	9-inch propeller clearance
Rigid static position	143 mm
10 degrees nose-up from static	59 mm
10 degrees, less 15 mm assumed deflection and 25 mm grass	19 mm
12 degrees, with those same deductions	About 2 mm

These are geometry cases, not measured deflection or a landing rating. At nominal balance the static distribution is approximately 13% nose wheel and 44% on each main wheel. Pitch, grass, rebound, side loads and uneven ground can consume the clearance.

For the 20 x 4 mm 6061-T6 main strip, elementary 3g bending gives about 90 MPa at 2.63 kg. Stress = wheel force x lever arm / section modulus; the lever arm is about 142 mm and strip section modulus 53.3 mm³. Specified stock yield is at least 240 MPa. This straight-beam comparison does not qualify bends, bolts, wood bearing or impact.

Assumed sink / stopping stroke	Vertical energy	Mean total reaction
1 m/s / 20 mm	1.32 J	About 3.6g
2 m/s / 20 mm	5.27 J	About 11.2g
2 m/s / 10 mm	5.27 J	About 21.4g

Vertical energy = 0.5 x mass x sink speed squared. Mean reaction = energy / stroke + weight. These scenarios explain why a 3g strip calculation is not a hard-landing proof. Horizontal flight energy is largely dissipated through aerodynamic braking and rollout, not the same vertical gear stroke.

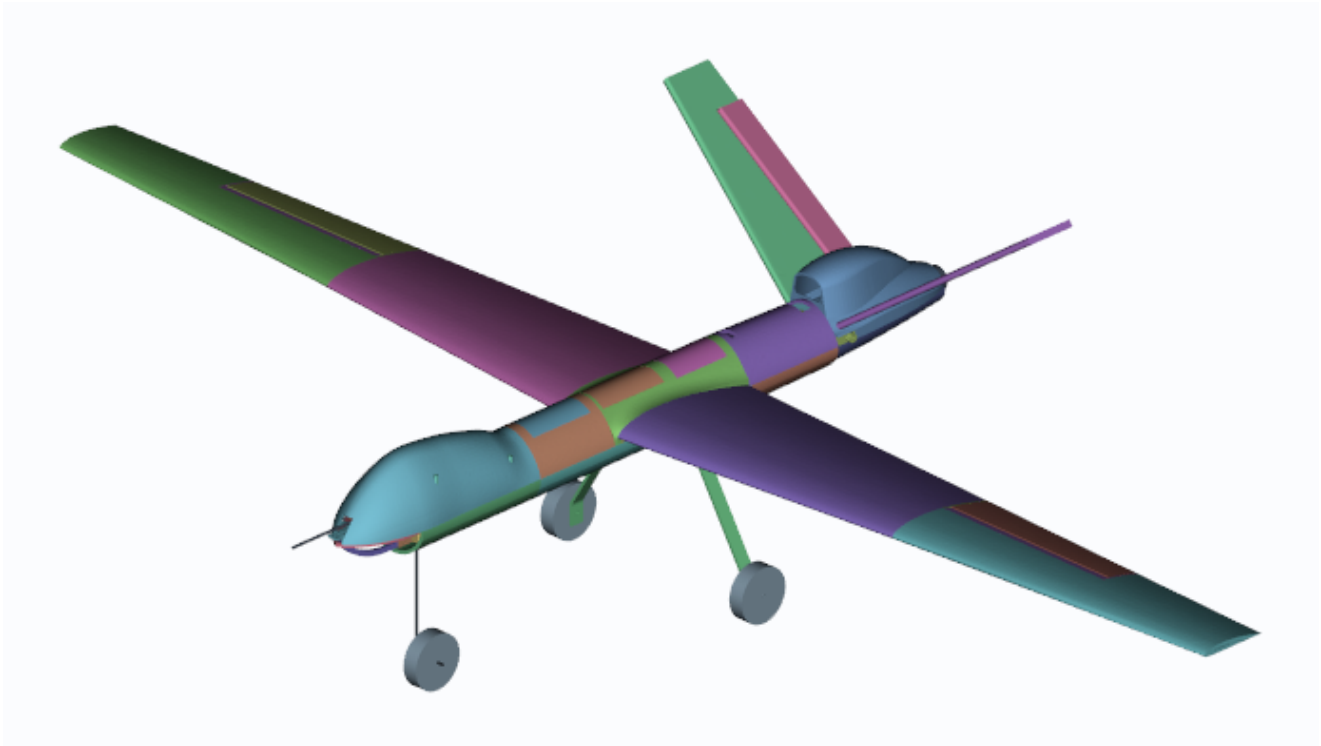
Before flight	Evidence to record
Mass and CG	Finished component weights, complete-aircraft balance, secured battery position and a justified flight CG range.
Controls and structure	Full combined control travel; loaded servo supply; joined wing/root and A12 tail bending/torsion checks; no residual damage.
Propulsion and launch	Actual current, thrust, voltage sag and temperature with selected propeller; demonstrated launch method and pilot recovery plan.
Systems and landing	Sensor calibration, radio range/failsafes, loaded compass interference, gear stroke, clearance and fastener retention.

Calculation sources

NASA lift/drag | NASA induced drag | TowerPro MG90S | Matek F405-WING V2. Geometry is from the supplied CAD. Material properties, aerodynamic coefficients and unweighed hardware are explicitly assumed. No ground or flight test is represented as completed.

01 / MQ9 airframe

Manufacturing Team / 2-metre RC model / Revision 2



Build the frame, body, mounts and controls with this guide. The Electronics Team guide covers wiring, sensor setup and calibration.

Build order	Finish each check before the next stage
1 / Coupons	Measure stock and test the cutting, printing and foam-forming processes.
2 / Skeleton	Dry-fit both wings, the fuselage and landing gear.
3 / Installation	Fit mounts, actual equipment, straps, controls and service openings.
4 / Covering	Form, fit and bond skins while retaining the specified access.
5 / Transfer	Record mass, fit and structural checks; hand over with the propeller removed.

Read the Weight and flight assessment chapter before the full batch. Digital geometry checks do not clear hand launch or flight; material, balance and propulsion checks remain.

02 / Shop files and stock

Folder	Workshop use
01_Plywood	CUT profiles by thickness, sheet layouts, quantities and fit coupons.
02_Foam	Developed patterns and forming references.
03_Print	Aircraft STLs, dimensions/quantities and printer coupon.
04_Hardware	Carbon schedule, metal gear geometry and forming references.
05_Assembly	Positioned STEP and assembly illustrations.
Print_Templates	Actual-size A4 cutting templates. Print at 100%, not Fit.

Machine work: laser/CNC plywood, print selected STLs, form and drill the metal gear. **Bench work:** knife-cut foam, form, bevel, glue, align, fasten and install linkages. The package does not remove these assembly steps.

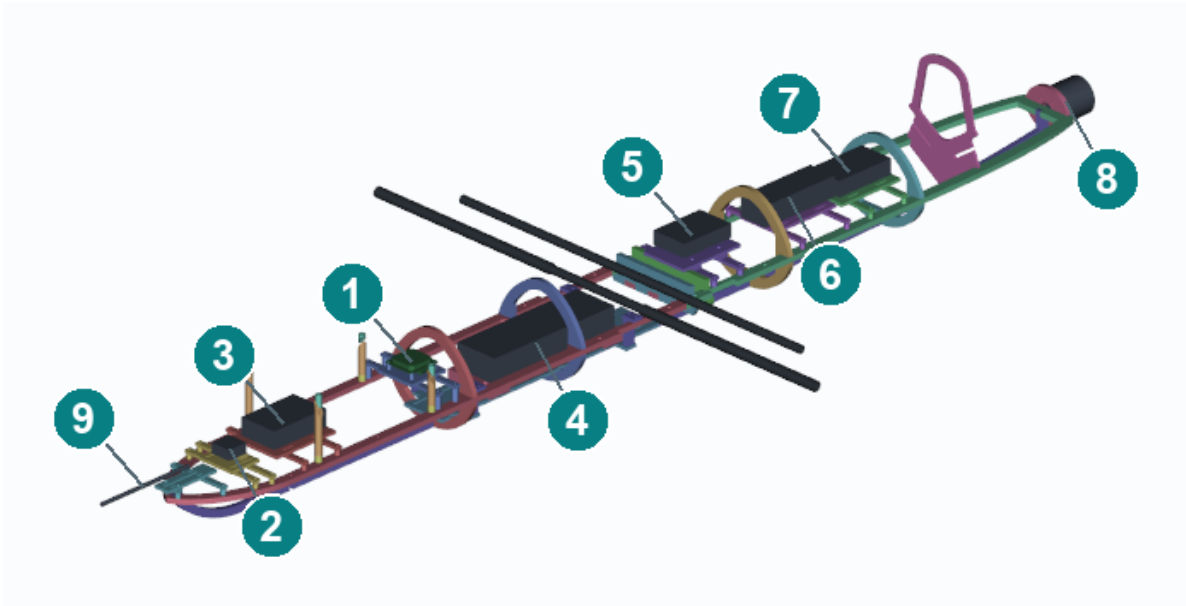
Use 5 mm FliteBoard Pro paper-faced XPS, 1000 x600 mm sheets. Use a knife or suitable CNC knife; laser processing requires the workshop to approve this exact material and extraction process.

Plywood: 24 pieces at nominal 5 mm, 6 at 3 mm and 7 at 2.5 mm. The two A44 ribs use the 4.990 mm geometry. Balsa is optional; it is not a substitute for the specified structural plywood.

The shop may re-nest 1220 x2440 mm layouts. Preserve scale, grain along long structural members and quantities. Do not scale a part to make a slot fit.

Tools: caliper, square, flat jig, knife, sanding blocks, clamps, drills/reamers, small nut drivers, long-nose pliers and a 1.5 mm ball-end hex driver. Make process coupons before cutting the aircraft set.

03 / Fuselage skeleton

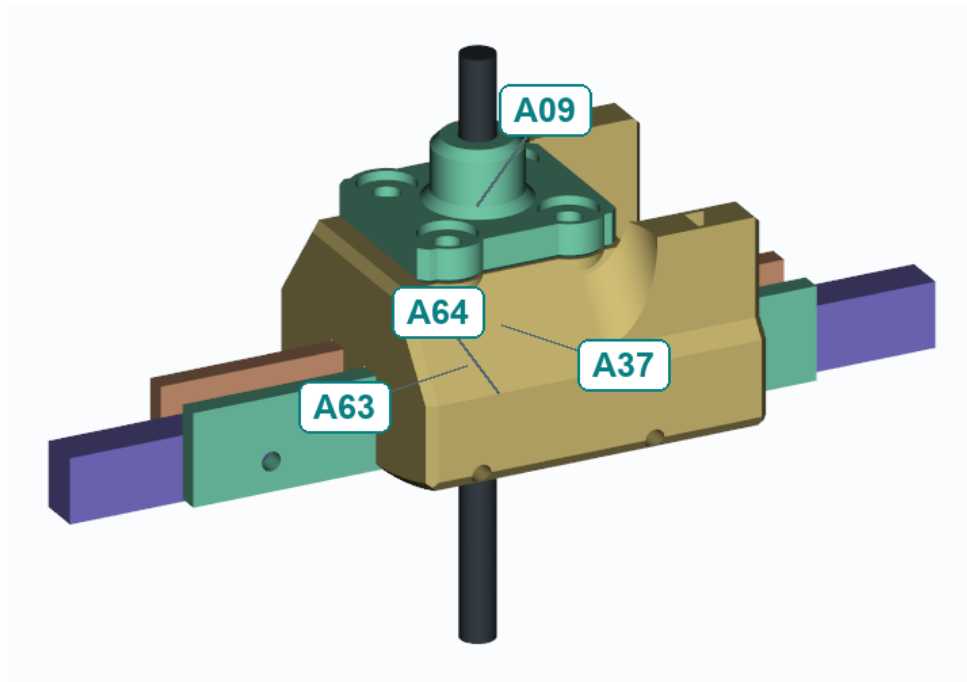


Assembly coordinates: X across the wings, Y toward the tail, Z up. Nose datum is Y=-439.65 mm. The STEP and mounting holes locate parts; this illustration shows equipment with covers removed.

1. Jig A62F/R on the centreline. Build their bolted nose bypass before bonding the frame.
2. Fit A13 and A43. Install A34 at Y=-130, A54 at Y=-5, A16 at Y=233.65 and A04 at Y=384.65. Seat the tabs fully without forcing.
3. Fit the main-gear box A25/A48 and bottom plate A47. Bond G03 below A47 with both M3 holes aligned. Fit tail bulkhead A18 and motor mount A26.
4. Dry-fit A59, A06, A12, both wings and all installation parts. Check equal left/right extents, square formers and straight centreline.
5. Bond after the complete dry-fit. Add modest fillets at gear, wing-root and motor joints. There is no separate A40 piece.

Keep the battery route open: the pack slides from the nose into its low central tray. Do not add a bulkhead, crossbar or wire bundle across this route.

04 / Nose gear and bypass

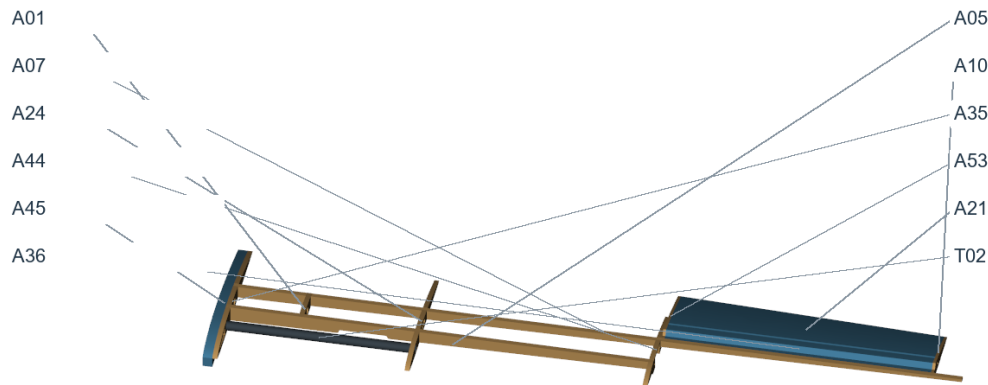


Below this bay, A63/A64 bridge the split spine around the steering shaft. Each cheek is 2.5 mm plywood, 72 x10.7 mm, with grain along its length.

1. Set A62F/R collinear with the **4.4 mm gap** centred at Y=-293.65. Fit the cheeks into A37, one per side of the 5 mm spine.
2. Install four transverse M2 bolts at Y=-320, -302, -282 and -264, Z=-37.964. Use washers and locking nuts. Inner bolts pass through A37; outer bolts clamp the wood stack.
3. Fasten A09 to A37. Fit the **4 mm spring-wire shaft** through the finished 4.2 mm bores. Fit the 4 mm clamp steering arm and retainers. Keep the shaft turning freely.
4. The wire reaches Z=+1. Trim only after checking full clamp engagement. Use mechanical retention, not glue. Fit the 76.2 mm nose wheel without squeezing its hub.
5. Keep the steering horn below the deck and check its full travel. Make the 6 mm skin clearance hole only after forming the lower nose skin. Recheck under load and steering deflection.

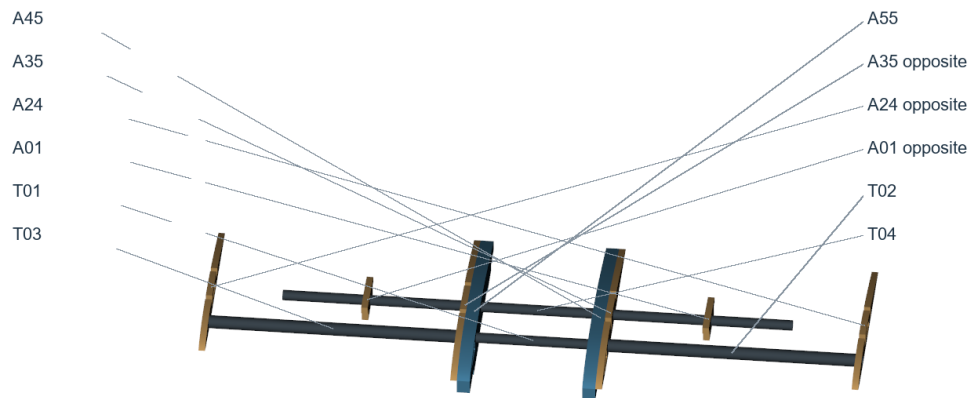
Colours identify parts; they do not specify paint or material. The cover, GPS and pitot attach independently of the steering shaft. Use nonmagnetic fasteners near the compass.

05 / Wing frames and controls



1. Build the right wing and its left-hand mirror. Fit A35, A24 and A44 to A05/A07. Fit A01 and the front sleeve. Use the part quantities, not the number of source-side examples.
2. Jig both wings straight and check equal incidence. Finish designated rib/end-plate edges to their assembly contour while preserving bores and slots.
3. Fit A53/A10, hinge rail A36 and aileron A21 on the right; A23/A11 on the left. Fit smooth 3 mm pins, retain both ends and leave about 1 mm axial freedom.
4. Mount an MG90S in each A44 opening using two M2 bolts, broad washers and locking nuts. The body opening is 23.4 x13.0 mm; verify the actual servo before cutting both ribs.
5. Set servos and controls neutral. Put each horn pin over its hinge axis and align its plane with the servo arm. Use H05B backing plates and through-fasteners. Add crush-resistant sleeves through foam or hollow printed controls.
6. Cut pushrods after fitting both rod ends. Fit retention clips. Check full combined travel and servo-arm sweep with the skin temporarily fitted; make a local service/rod exit opening only where needed.

06 / Joiners and wing retention



Tube	Section and length	Assembly position
T01	8 OD /6 ID; 406 mm	X=-203..203; axis Y60.454, Z15.632
T02/T03	10 OD /8.2 nominal ID; 199.35 mm each	Right X53..252.35; left mirror
T04	8 OD /6 ID; 390 mm	X=-195..195; axis Y100.500, Z15.070

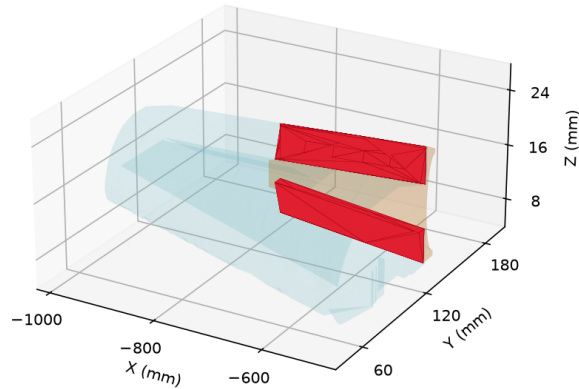
Use a measured matched sliding pair. The 5 mm solid carbon rods are spare material, not the specified joiners. Do not force a tube or sand away its structural wall.

Centre both joiners and test both wings together. Bond sleeves only, keeping adhesive off the sliding joiner. Mark the insertion positions; front engagement is about 150 mm per wing.

Fit three M2 root retainers per wing with washers and locking nuts. Check twist, spanwise escape, wiring disconnects and hinge clearance. Label the left/right servo connectors and strain-relieve them.

07 / Forming the foam

Wing hinge inner core | red = CAD interference to relieve

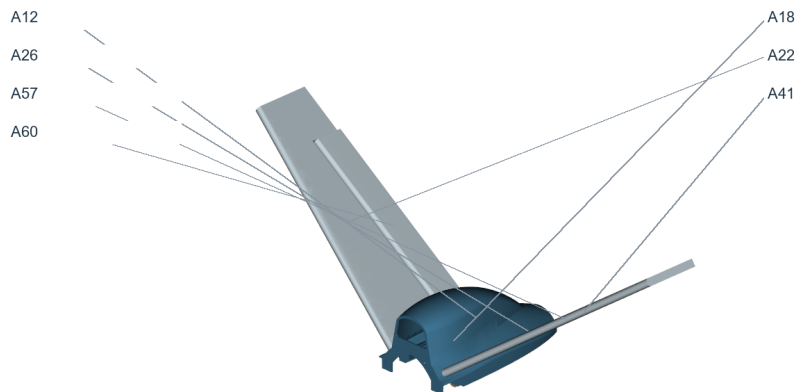


Blue: A27 foam. Tan: A23. Red: trim contact region. View cropped; axes not equal scale.

1. Print at Actual Size /100%. Measure the 100 x20 mm rectangle. Join only matching tile marks with the specified overlap.
2. Mark OUTSIDE and the F-series ID. Keep the outside paper as the developed pattern reference. Test the tightest curve on scrap first.
3. Peel inside paper only in curved zones. Relieve/compress the core gradually over the actual frame. Do not stretch a short edge to close a seam.
4. Bevel inside edges to the actual meeting angle. Butt outside-paper edges; use separate 15 mm internal paper straps. Keep glue and straps clear of moving parts.
5. Dry-fit a whole wing and its control before bonding. Hold alignment through cure. Use adhesive approved for XPS; ordinary CA must not reach exposed foam.

CUT lines are finished skin edges. F23/F24 alone have a 1 mm paper-only lower lip. F31/F32 each need two layers and have 5 mm sacrificial shaping stock. Other panels have no added lap or trim.

08 / Tail and service openings



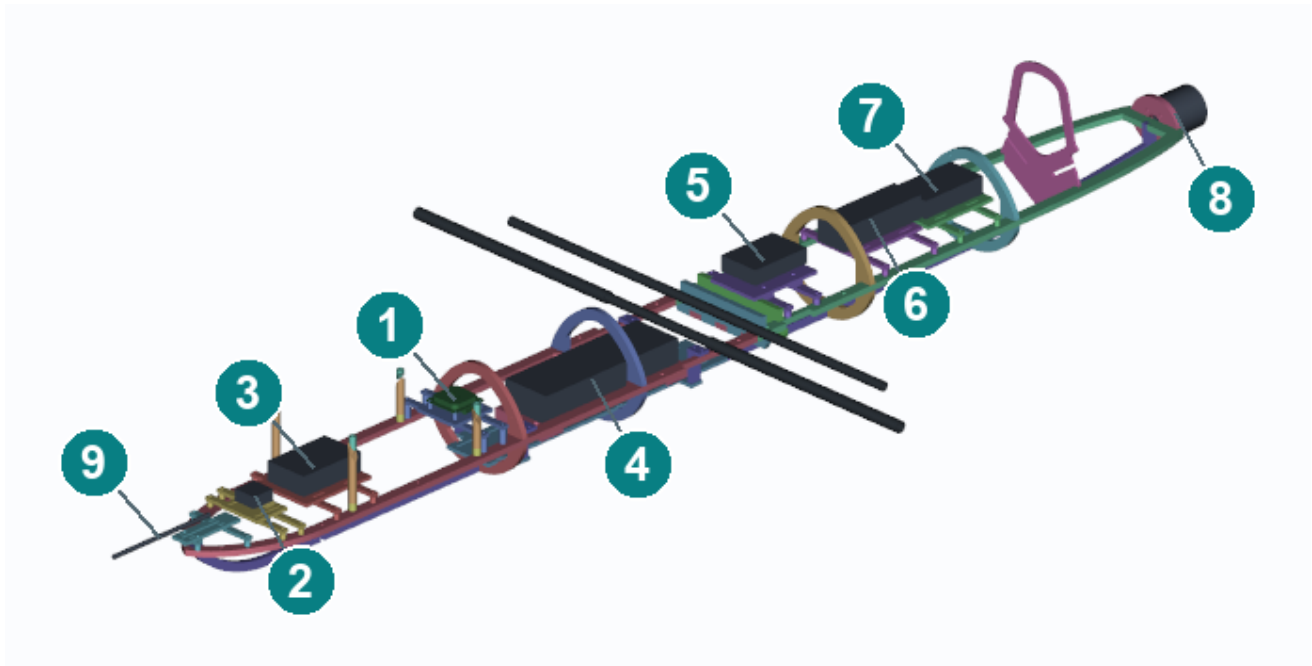
Form the fixed tails with the split collars. Check their fit in A12 and equal angles before bonding. Laminate each moving tail from two 5 mm layers, trim to the finished outline and shape long edges to R5.

Fit both MG90S servos in A12 using the slotted ear holes. Tape-hinge the controls on the indicated line. Fit backed horns and adjustable pushrods; check combined pitch/yaw travel without binding.

Location	Finish after forming
F25/F26 forward roof	Join the two skins, then make the 50 x100 mm plan-view service opening: X +/-25, Y=-115..-15. Retain the lid with paper edge supports and replaceable reinforced tape. Keep A54 intact.
F23/F24 rear roof	Join as one removable cover. Add 15 mm internal paper supports and six reinforced-tape retention tabs. Do not glue this cover permanently shut.
Cooling	Inlet: 24 x12 mm at Y250. Outlet: 30 x20 mm at Y374. Add a small forward-facing scrap-foam scoop over the inlet to direct air toward the ESC. Verify cooling by test.
Gear exits	Form skins first. Mark actual gear crossings; provide 3 mm nominal clearance around the main legs and a 6 mm nose-shaft hole. Reinforce the paper edges and verify loaded clearance.

These openings are finishing operations on the formed skins. Do not use their plan-view dimensions as replacement flat foam patterns.

09 / Equipment mounting map

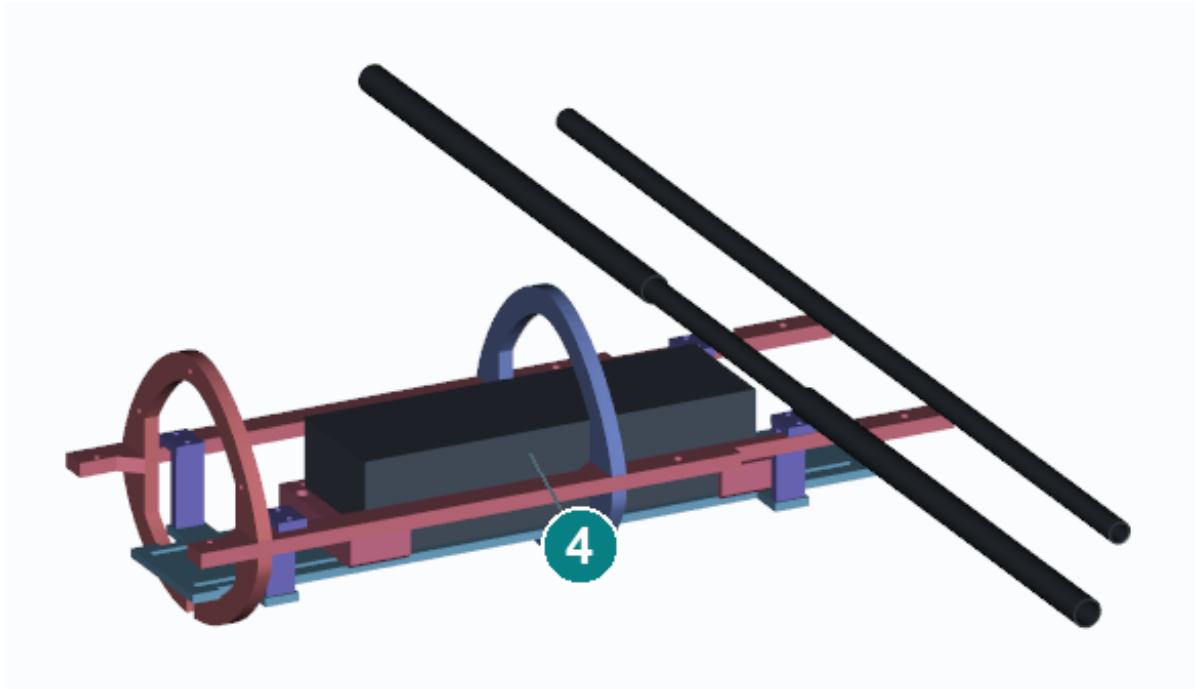


Mark	Equipment	Station from nose / mounting
1	M9N-5883	286.65 mm; I09 inside nose on A13
2	ASPD-4525	76.65 mm; I13 on A13
3	FS-iA10B	139.65 mm; I10 on A13
4	GenX battery	Centre adjustable 379.65..459.65 mm; I01 below deck
5	F405-WING V2	609.65 mm; I04, 30.5 mm grid
6	40 A ESC	719.65 mm; I12 on A43
7	SiK air radio	791.65 mm; I11 on A43
8	Motor	A26 mounting face at 1022.30 mm
9	Pitot	I14; pressure ports start at least 50 mm ahead of the nose

Install the battery with main and balance leads exiting toward the tail. Keep nearby power wiring aft of Y=-37, at least 100 mm behind the GPS module. Never route a battery lead across the internal GPS bridge.

Equipment shapes in illustrations are placement references. Use the actual connectors and secure every item with the specified fasteners/straps. Route wires along the frame sides, away from the battery insertion path.

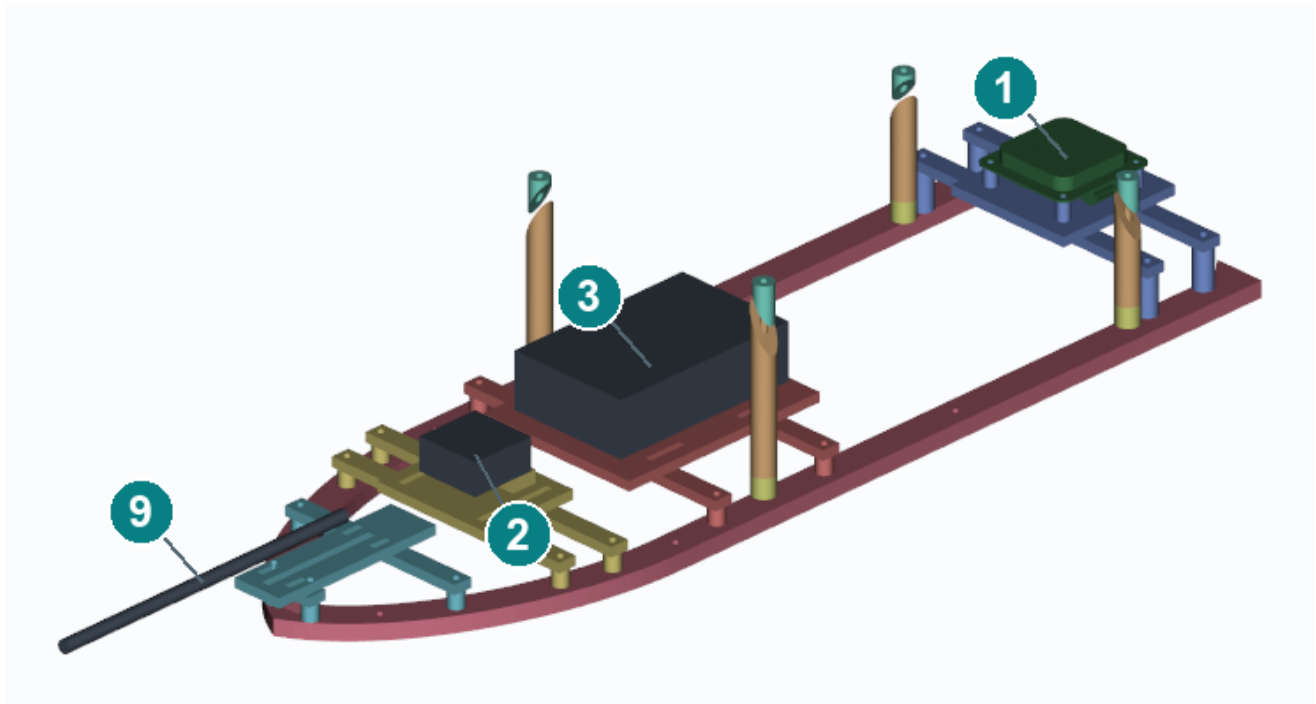
10 / Battery tray and access



1. Bolt four I17 hangers to A13 at X +/-40, Y=-115 and 65. Bolt I01 below them at X +/-32. Tray underside is Z=-28. Its local edge notches clear A34/A54; do not cut those formers.
2. Fit a 1 mm non-slip pad. The 138 x43 x33 mm pack sits with its bottom at Z=-24. Confirm clearance to both tubes with the actual pack; no wiring may pass over the pack in this gap.
3. Fit I02 corner stops with their fingers toward the pack. Their centres are 73 mm ahead of and behind the selected battery centre. Use M2 screws, washers and locking nuts in the slots.
4. Remove the front stop to load. With the nose cover removed, place the pack forward of A34 and slide it aft into the tray. Fit the front stop and two separate 20 mm straps around both pack and tray. Keep stops padded; never squeeze the pouch.
5. Set the initial location during dry assembly, then balance the finished aircraft. Available centre travel is Y=-60..+20. Use a ball-end hex driver and long-nose pliers for fasteners under a joiner; verify access before closing skins.

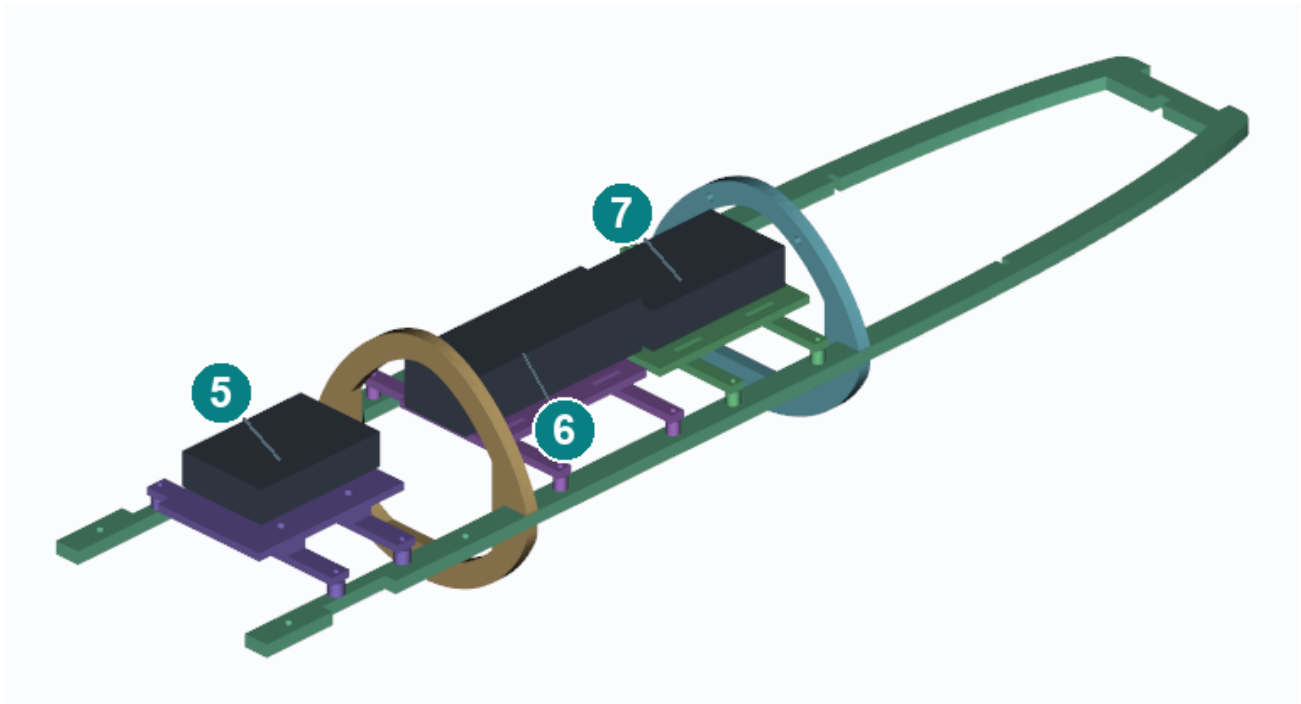
The F25/F26 opening and front A06 hatch give access to straps and stops. Support tabs carry the printed hatch; two replaceable 20 mm reinforced-tape strips retain it.

11 / Nose sensors and cover



1. Bolt I14 to A13. Fasten the supplied pitot tab in the centre slot with an M2 bolt, washers and locking nut; add the anti-rotation strap. Align the tube forward, parallel to the fuselage centreline.
2. Start with every pressure port at least 50 mm ahead of the nose. Keep ports clear of straps and adhesive. Connect both tubes to I13 without kinks; identify total and static pressure ports from the sensor/kit markings.
3. Fit ASPD-4525 with two M2 screws on its 15 mm hole spacing. Insulate the board. Keep the pressure tubing and I2C cable accessible under the nose cover.
4. Bolt I09 to A13 at Y=-153 inside the nose, using four M2 nylon feet. Mount M9N-5883 level, antenna up and arrow forward on its 26 mm grid. Keep bolt tips above Z11. Route battery power leads aft; use nylon fittings nearby and check compass interference with the motor running.
5. Fit I10 and strap the receiver to its insulating pad. Support the two active antenna tips independently at right angles, clear of carbon and power wiring. Check range with the installation complete.
6. I00 six-mm spacers support the four I05 columns; Compact I06 washers match their curved cover seats. Use nylon cover bolts near the compass. Front columns are at Y=-300, rear at Y=-170. Remove four cover bolts and lift the cover. GPS, receiver and pitot remain on the frame; keep their wires and tubes clear.

12 / FC, ESC and telemetry



1. Bolt I04 to A43. Mount the F405-WING V2 on its 30.5 mm grid with 6 mm nylon spacers and supplied grommets. Do not crush the grommets. Keep USB, SD and connectors accessible through the rear A06 hatch.

2. Strap the ESC to I12 and the air radio to I11 over insulating pads. Leave connectors and the ESC cooling face clear. Straps must restrain the cases without loading a PCB connector.

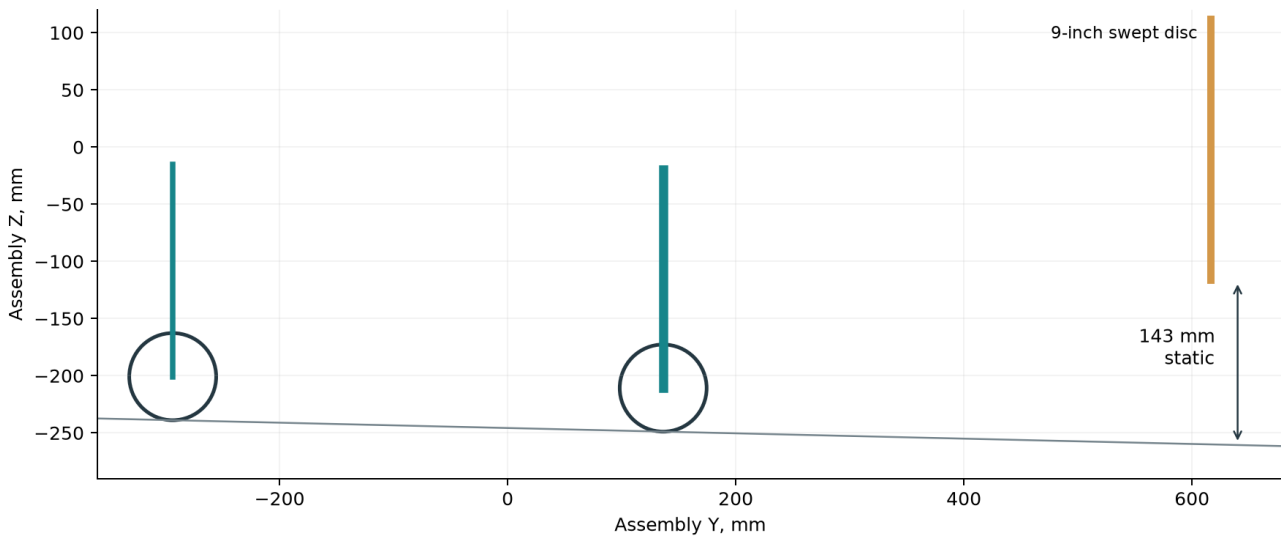
3. Keep the FC-to-ESC battery-power run short. Route motor phase extensions aft along the frame side, away from radio/sensor cables. Use appropriate flexible copper wire and support every connector; do not extend battery leads casually.

4. Support the 433 MHz antenna separately from its connector. Fit the correct antennas to both radio units before power. The purchase listing conflicts on power; use the actual radio label and firmware limits, with 100 mW as the requested setup.

5. Fit the removable F23/F24 cover and cooling inlet/outlet. Verify wire clearance, connector access and measured ESC temperature under load.

The Electronics Team guide gives pad-to-pad wiring, BEC isolation, output assignments and calibration. The four rear-bay carrier foot holes are already in A43 for each component.

13 / Landing gear for grass



G01: 20 x4 mm 6061-T6 strip, 12 mm inside bend radius. Use the formed STEP as the dimensional authority. The K=0.40 flat DXF is a tooling starting point: qualify a bend coupon and adjust allowance before cutting the final blank. Reject cracks; do not soften the part without restoring and verifying the required material condition.

Bolt G01 below G03/A47 with two M3 through-bolts and broad washers. Use two smooth-shank M4 wheel axles. Fit three lightweight 76.2 mm wheels with 4 mm bores; adapt spacers to measured hubs.

G02 uses 4 mm spring wire formed by a suitable shop. Nose wheel reference: 22 mm hub width, centre X26. Allow for the inner collar/washer, wheel, outer washer/collar and positive retention on the 46 mm axle.

9-inch propeller clearance	Calculated rigid geometry
Static, flat ground	143 mm
10 degrees nose-up from static	59 mm
Same 10-degree case, minus assumed 15 mm deflection and 25 mm grass	19 mm remaining

These are design cases, not measured gear deflection or a landing rating. Check the assembled gear at measured aircraft mass, including fore-aft loads, joint motion, wheel freedom and skin clearance. Use short, even grass; the figures do not cover rough ground or arbitrary flare angles.

14 / Prints, motor and finishing

Group	Starting process; qualify actual material
A06/A59 shells and access lids	Available PLA+ is not LW-PLA. Start 0.2 mm layers, 2 perimeters and 3-5% infill; inspect thin walls and record slicer/finished mass.
A12 structural tail housing	Carries tail-root and servo loads. Qualify local slot walls, layer orientation, root bonds and the A12-to-A18 connection. The 2-wall/5% slicing case is a mass estimate, not an accepted production recipe.
Ailerons and hinge rails	Start 3 perimeters, 10% infill. Keep supports out of bores; fit smooth pins and verify full motion.
Load-bearing mounts and carriers	Start 4 perimeters, 30% infill, with solid material around bolts. Orient loads to avoid splitting layer interfaces. Test the actual material/process.

Print 42 installed pieces from 38 STL files, using the quantity table. The fit coupon is extra. Files are millimetres at 100%; some long parts need a larger printer. Do not make an arbitrary split or rescale.

Clear all passages, measure the coupon and inspect layer bonds. Record every finished print mass. Slicer settings alone do not establish strength or heat resistance.

A26 motor mount: four 3.2 mm holes on a 19 mm pitch circle; adjacent centres 13.435 mm. Preserve the 12 mm centre opening. A62R ends flush with the rear mounting face at Y582.65; no spine tab extends into the motor. Measure screw engagement so no screw can reach windings.

Use the selected Emax 2807 / 9-inch three-blade / 4S /40 A combination as the design assumption. Current, thrust and temperature are still measured before flight. Keep the propeller removed during assembly and configuration.

Body assembly still needs foam forming, adhesive, fasteners and linkage adjustment after the machine work. Weigh the finished body before the Electronics Team chooses the battery position.

15 / Body acceptance and transfer

Check	Record on the actual aircraft
Stock and scale	Measured ply/foam thickness; slot coupon; 100 x20 mm rectangle; printer dimensions.
Structure	All 37 wood pieces present; correct grain; bonded joints and bypass fasteners; straight frame and matched wings.
Controls	Five MG90S secure; retained pins/rod ends; backed horns; full combined motion without binding or servo stall.
Battery and access	Actual pack fits and can be removed; both straps and stops restrain it; tubes and cables remain clear.
Gear	Material/process accepted by shop; wheels and steering free; load/deflection and attachment checks recorded.
Skins	No forced stretching, open seams or twist; reinforced gear exits and removable covers secure.
Electronics interfaces	Pull strings, labelled wing connectors, pitot tubes, independent antenna support and service access.
Mass	Body and individual print masses recorded; include glue, fasteners, wiring and finish.

Balance reference: 522.3 mm aft of the nose, or 22.2 mm aft of the front joiner, is the geometric quarter-MAC. It is not an approved flight CG range. Use the measured mass worksheet and final complete-aircraft balance.

Battery movement changes CG by battery mass / total mass times its travel. The available 80 mm travel is useful trim; do not assume it can correct any weight distribution.

Before hand launch, the Electronics Team and an experienced fixed-wing pilot must review the finished mass, CG, control authority, propulsion and launch procedure. Complete the prop-removed launch-detection, range and failsafe tests first.

Electronics build guide

Matek F405-WING V2 / ArduPlane / FlySky / five servos with nose steering

For the **Electronics Team**. Follow the steps in order and record actual readings in the build record.

Remove the propeller for wiring, firmware, calibration, servo and failsafe work. Refit it only for a separate controlled propulsion test after bench checks pass.

Equipment	Build assumption
Flight controller	Matek F405-WING V2; not V1, WTE or TE
Navigation	Matek M9N-5883 GNSS + compass; 32 x 32 x 10 mm, 14.5 g
Controls	Five TowerPro MG90S: two ailerons, two V-tail controls, nose steering. Nominal 13.4 g each.
Power	GenX 4S 5200 mAh: 138 x 43 x 33 mm, 512 g. ReadytoSky 40 A 2-4S with 5 V / 3 A BEC.
Propulsion	Emax ECO II 2807 1300 KV + Gemfan 9045 CL three-blade; selected 4S /40 A setup
Radio / airspeed	FlySky FS-i6X / FS-iA10B; Matek ASPD-4525; SiK 433 MHz telemetry.

Work in this order

Identify parts -> build and meter harnesses -> configure firmware -> connect radio and sensors -> centre and test servos -> test failsafes -> install and balance -> qualify propulsion. Save a parameter file after each successful stage.

Five servos: two ailerons, two tail controls and one nose steering. Establish control throws, flight CG, speeds, gains and return altitude from the completed aircraft and measured tests.

Match the confirmed hardware

Connector shape and wire colour are not a pinout

Component	Connection and check
M9N-5883	5V to FC 5V; G to G. Module TX to RX3, RX to TX3. Compass DA to DA2, CL to CL2. Mount level with arrow forward; use manufacturer ArduPilot orientation and verify heading. [18]
ASPD-4525	5V to FC 5V, SCL to CL2, SDA to DA2, G to G. ARSPD_TYPE=1; ARSPD_BUS=1. GPS compass and airspeed share I2C2 in parallel. [1,6]
ESC signal plug	This ReadytoSky listing specifies a 5 V / 3 A BEC. Depin and insulate its BEC output lead; retain signal and signal ground. Verify the actual plug before powering. Do not join it to the FC servo or 5 V rail.
MG90S x5	Use the FC Vx 5 V setting and initial 50 Hz PWM; do not select 6 or 7.2 V. Centre and limit each servo before attaching linkages. Published stall torque is not a continuous load rating. Load-test the whole rail. [19]
Battery / connectors	Design pack: 138 x 43 x 33 mm, 512 g. Meter connector polarity and record charger limits. Never connect the 4S main lead to a receiver, GPS, servo or 5 V pad.

Make a connector record

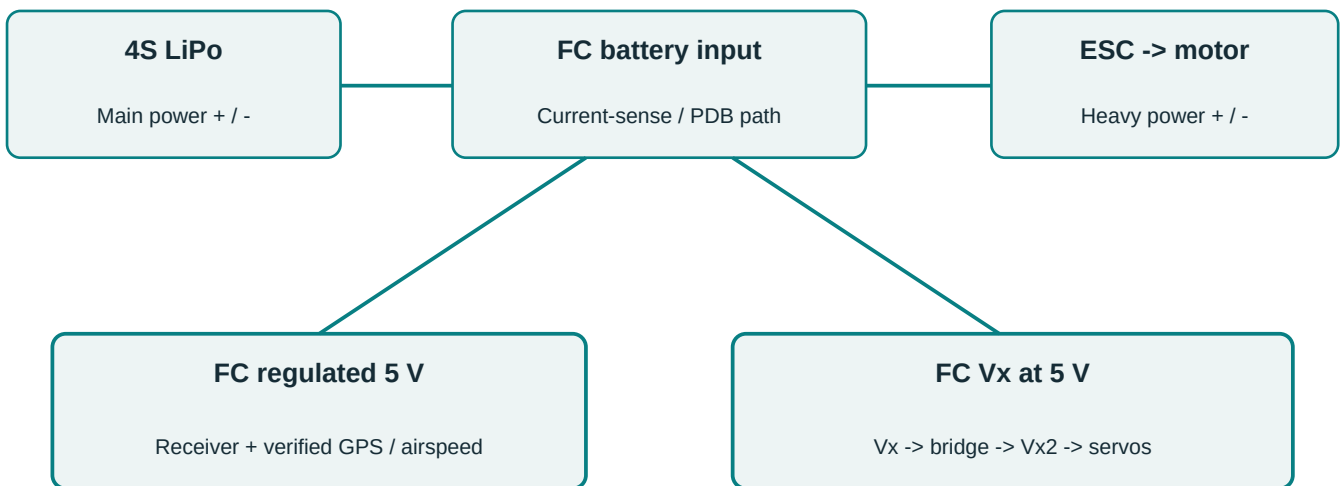
Draw each plug from a stated face. Number its cavities and label signals. With power removed, continuity-check both ends and label the harness.

Pad names below identify functions. Verify connector cavity numbers from the exact hardware manual before soldering.

Power architecture

One positive source per rail; common signal ground

Route battery and ESC power through the FC power-distribution pads. Match the V2 board layout [1].



ESC signal + ground -> FC S1 + G

ESC BEC output (if present) -> insulated; never joined to FC rails

Build and meter

1. Fit strain-relieved battery and ESC power leads to the documented pads. Keep high-current loops short. Select connectors and wire for measured load; insulate exposed joints.
2. Keep electronics 5 V and servo Vx positives separate. S5-S9 use Vx2: bridge Vx to Vx2 for this internal-BEC plan. Check before plugging in servos. [1]
3. If tests require an external BEC: remove the bridge, verify isolation, power Vx2 from that BEC and move all servo positives to that rail. Retain grounds; never join two BEC positives.
4. With peripherals unplugged, check polarity and hard shorts. First power through a suitable current limiter, then meter 5 V and Vx/Vx2 at far connectors. A limiter may brown out under servo load; it does not qualify flight power.

Wire the signal harness

FC outputs and transmitter channels are independent

FC connection	Destination / setting
S1 + G	ESC PWM signal + ground. SERVO1_FUNCTION=70 (Throttle). Isolate BEC output lead if present.
S2	Unused, SERVO2_FUNCTION=0; leave empty.
S3 + Vx + G	Left aileron. SERVO3_FUNCTION=4 (Aileron).
S4 + Vx + G	Right aileron. SERVO4_FUNCTION=4 (Aileron).
S5 + Vx2 + G	Left tail. Initially SERVO5_FUNCTION=79 (V-tail left). Use the V-tail direction checks; physical motion governs.
S6 + Vx2 + G	Right tail. Initially SERVO6_FUNCTION=80 (V-tail right). Use the V-tail direction checks.
S7 + Vx2 + G	Nose steering A39: SERVO7_FUNCTION=26 (GroundSteering). Omit only with a deliberately fixed nosewheel.
S8-S9 / LED	Unused; disconnect. Check SERVO8-10_FUNCTION for unwanted assignments.
RX2 + 5 V + G	FS-iA10B iBUS servo-output signal + supply + ground. Not SENS telemetry port. BRD_ALT_CONFIG=0.
RX3 / TX3 / supply / G	M9N-5883: module TX -> RX3, RX -> TX3; 5V -> 5V; G -> G. SERIAL3_PROTOCOL=5 (GPS).
DA2 / CL2 / supply / G	M9N compass DA + ASPD SDA -> DA2; both CL/SCL -> CL2. Both on regulated 5 V and G. Check both devices detected after connection.

Custom output map: write and read back every assignment before connecting the ESC. S1 is throttle only after configuration. [2-4]

Start with supported 50 Hz PWM. Use conventional PWM for this ESC; leave DShot disabled. Check shared timer groups before changing rates. [2]

Firmware and first connection

Mission Planner on Windows; menu names can change

Install the matching target

1. Confirm **F405-WING V2**. Connect a data USB cable with battery, ESC and servos disconnected.
2. Install stable **ArduPlane / MatekF405-Wing**, version 4.4 or newer. Do not select Copter, INAV, F405-STD or F405-TE. [1]
3. With an ArduPilot bootloader, load the matching .apj. Otherwise follow the board DFU first-install procedure using firmware with its bootloader. [2]
4. Record target and version. Save initial parameters and inspect sensor messages. Insert a supported logging microSD with power off.

Configure before powering outputs

Configure: write page 4 settings in Full Parameter Tree/List. Reboot and read back. Use Mandatory Hardware for sensors, radio and modes; Servo Output for servos.

Do not bulk-import an internet parameter file. Keep arming checks enabled. Resolve each failed check instead of setting `ARMING_CHECK=0`. Save the parameters actually read from this board.

Recovery route

Recovery: disconnect propulsion power. Reconnect USB and restore the last known parameter file. Repeat sensor, output and failsafe checks after firmware changes.

Nose steering: centre A39 with `SERVO7_TRIM`; set safe endpoints and reversal so right rudder steers right. Push-test with motor off. Set `GROUND_STEER_ALT` deliberately. Stabilized steering needs healthy heading and `STEER2SRV_P` based on measured turning diameter. [14]

Radio input and modes

Plain controls from FS-i6X; mixing inside ArduPlane

Create an airplane model; select AFHDS 2A and bind the receiver. Disable transmitter V-tail/elevon mixes and centre trims. Use the receiver iBUS servo port with regulated 5 V. [5]

Input	Intended function
CH1	Roll; RCMAP_ROLL=1
CH2	Pitch; RCMAP_PITCH=2
CH3	Throttle; RCMAP_THROTTLE=3
CH4	Yaw; RCMAP_YAW=4
CH5	Three-position mode switch; FLTMODE_CH=5. Measure its actual detent values.
CH6 optional	Dedicated arm/disarm only after verifying the installed firmware option and behaviour; never share with mode control.

Calibrate

Calibrate inputs: power RX from verified FC 5 V. Check each stick moves the intended bar. Sweep sticks and switches; finish at low throttle with other sticks centred. Save. [8]

Input directions: right roll, pitch up, more throttle and right yaw must increase their normalized inputs. Correct RCn_REVERSED first; adjust surface direction later with SERVOx_REVERSED. Recalibrate after endpoint changes. [8]

Modes: initially set all six slots MANUAL. Assign the middle detent to FBWA. Keep the third MANUAL until navigation checks pass, then deliberately assign RTL. Check HUD mode at every detent and transition.

Receiver telemetry is not automatically flight-battery voltage. Keep antenna active ends clear of carbon and power wiring; range-test after final installation.

Sensors and battery readings

Calibrate in final orientation; inspect the actual response

Attitude, heading and position

Attitude: secure FC and set AHRS_ORIENTATION. Calibrate accelerometers in six positions, then level against the aircraft datum. Verify HUD roll/pitch follow the aircraft.

Compass: use the M9N-5883 external compass. Mount level and arrow forward, at least 100 mm from power wires, ESC, motor and ferrous parts. Manufacturer ArduPilot setting: Rotation None. Calibrate and verify known headings; confirm firmware detects the fitted QMC5883L/P. [18]

GPS: use SERIAL3_PROTOCOL=5 and the appropriate GPS type. Check outdoors for plausible position, fix and time. GPS-only hardware does not provide magnetic heading.

Battery monitor - V2 values

Parameter	Initial value
BATT_MONITOR / voltage / current pins	4 / BATT_VOLT_PIN=10 / BATT_CURR_PIN=11
BATT_VOLT_MULT / BATT_AMP_PERVLT	11.0 / 66.7; older-board 31.7 is wrong for V2
BATT_CAPACITY	5200 mAh nominal; validate usable capacity separately

Calibrate against a meter:

New voltage multiplier = old multiplier x meter voltage / reported voltage.

New current scale = old scale x reference current / reported current.

Use a steady meaningful current, not near zero. Check idle offset and consumed capacity. [1,4]

Set low/critical thresholds, delays, reserves and actions from pack specifications and measured loaded behaviour. Record them; the C-rating does not prove endurance.

Servos and V-tail direction

Left/right viewed from behind the aircraft, facing forward

Centre servos electrically before fitting horns. Align linkages mechanically; use SERVOx_TRIM for small corrections. Set conservative endpoints and check every hinge through full travel.

Test	Required response
MANUAL: roll right	Right aileron trailing edge up, left down; reverse for roll left.
MANUAL: pitch up	Both tail trailing edges move in upward pitch-producing sense; reverse for pitch down.
MANUAL: yaw right	Both tail trailing edges move toward aircraft right; reverse for yaw left.
FBWA, centred sticks: tilt aircraft right	Left aileron up, right down: correction opposes tilt.
FBWA: raise nose	Tail command produces nose-down correction; lowering nose produces nose-up correction.

V-tail correction: wrong in both pitch and yaw â†’ reverse output. Wrong in only one axis â†’ exchange functions 79/80 and repeat both tests. Physical motion governs the final mapping. [9]

Roll-to-rudder testing requires nonzero KFF_RDDRMIX; restore its intended value afterward. A stationary yaw test does not prove heading hold. [9]

Combined commands and supply loading

Test all four full pitch/yaw combinations. Check binding, buckling, buzzing and saturation. Adjust mechanics, endpoints and MIXING_GAIN; repeat MANUAL and FBWA tests. [9]

Load-test all servos together without prolonged stall. Capture voltage dips, current peaks, temperature and resets. A multimeter may miss brief dips. Limits: Vx 5 A continuous / 6 A peak; electronics 5 V rail 2 A. Verify margin. [1]

Airspeed and propulsion

Sensor response is not calibration; motor spin is not thrust

Airspeed branch

Use ASPD-4525 on I2C2: ARSPD_TYPE=1, ARSPD_BUS=1. Reboot and confirm airspeed health and compass detection. [1,6]

Put pitot in clean airflow, clear of prop wash. Match dynamic/static ports; avoid leaks and kinks. Let temperature settle. Shield from wind without sealing ports; run Preflight Calibration for zero. [10]

Apply gentle airflow: reading should rise and return near zero. Avoid pressure, liquid or blowing into tubing. Start ARSPD_USE=0; some Plane versions still use a healthy sensor for speed scaling. [10,11]

If readings are suspect, set ARSPD_TYPE=0 and resolve them. Ratio calibration needs a safely flying aircraft and log review. Bench airflow does not establish true airspeed or stall speed.

ESC and motor

Motor mount: four M3 holes on 19 mm PCD; adjacent spacing 13.435 mm. Check actual motor fit and safe screw engagement. The 9-inch /4S /40 A setup is the selected design assumption. Record current, thrust and temperature before flight. [15]

Confirm ESC PWM range and calibration method. FC/RX must stay powered while ESC power switches separately. Use a documented isolated setup or servo tester; do not backfeed power rails. [12]

If the ESC manual specifies high/low calibration: remove prop, switch ESC off, command maximum, power ESC, wait for its indication, then command minimum. Verify acceptance. Do not use multicopter calibration for Plane. [12]

Check stopped motor when disarmed and at minimum. Restrain the airframe for a brief low-speed rotation check. Power off before swapping two phase wires to reverse. Confirm pusher prop orientation produces forward thrust.

Failsafes: prove the trigger

Prop removed; disconnect motor power for configuration tests

With prop removed and telemetry recording, switch the transmitter off. Observe RC inputs, failsafe message, mode and recovery when the link returns.

Radio loss

Set `THR_FAILSAFE=1`. Check whether RX stops frames, flags loss or holds/presets channels. Held normal channels may hide link loss. Configure a detectable condition and repeat. [13]

For low-throttle detection, put `THR_FS_VALUE` between measured loss and valid-low throttle. Example only: loss 900, valid low 1000, threshold 950 us. Require clear separation.

Candidate policy: short loss `CIRCLE`, `FS_SHORT_ACTN=1`; sustained loss `RTL`, `FS_LONG_ACTN=1`. Verify firmware labels and both timeouts. Set valid home, return height, terrain clearance and power reserve before flight. [13]

Battery and navigation failures

Record low/critical voltage, capacity, delay and actions. Test with a suitable variable supply or temporary thresholds, then restore and read back flight values. Never deeply discharge the LiPo for a test.

A ground `RTL` mode change does not prove a safe return. Test navigation-failure policy in simulation, then controlled evaluation with an experienced pilot. Do not unplug live sensors near a running prop.

Record trigger, delay, action and recovery. Repeat from every intended flight mode. If a failsafe cannot be observed or explained, keep the flight gate closed.

Installation and release record

Complete on the actual aircraft

Restrain the adjustable battery. Separate power wiring from GPS, compass and receiver. Protect wires at edges, strain-relieve plugs and label wing/tail connectors. Keep USB, SD, rail bridge and battery access open.

Provisional setup balance station

Provisional balance mark: 522 mm aft of nose, or 22.2 mm aft of front joiner. This is geometric quarter-MAC, not an approved flight CG range. Check complete mass, control authority and final CG with an experienced fixed-wing pilot.

xCG = sum(mass x position) / total mass

Battery adjustment: battery shift = total mass / battery mass x desired CG shift.

Use one nose datum, aft positive. Include wiring, glue and finish. Mark repeatable pack positions; verify assembled balance and lateral balance.

Record	Measured / approved value
Firmware / saved parameters	_____
5 V minimum / Vx minimum / servo peak current	_____
Voltage calibration / current reference	_____
RC loss trigger / short / long / recovery	_____
Battery low / critical / actions	_____
Total mass / battery mass / actual CG from nose	_____
Desired CG shift / required battery movement	_____
Control throws / propulsion test / flight CG review	_____

Before flight: verify connectors, loaded supplies, control directions, range/failsafes, sensors, structure, CG and propulsion. Measure current, thrust, voltage and temperature on a proper restrained test fixture.

SiK telemetry / UART1

433 MHz, stated 100 mW / air radio to FC; USB radio to laptop

The retailer mixes 100 mW and 500 mW specifications. Use the actual radio label and firmware limits. A power setting does not identify the hardware. Fit the correct antennas before applying power.

Air radio function	F405-WING V2 connection
Supply / ground	Regulated 5V / G. Never VBAT or the servo Vx rail.
Radio TX	FC RX1
Radio RX	FC TX1
CTS / RTS, if present	Leave disconnected for the initial no-flow-control wiring; disable flow control if that parameter is exposed.
Antenna	Correct 433 MHz antenna; support it independently so the connector carries no bending load.

Connect and configure

1. With power off, map the supplied cable by signal labels and continuity. A Pixhawk plug is not a Matek pinout. Do not use cable colour as proof.
2. Set SERIAL1_PROTOCOL=2 for MAVLink2 and SERIAL1_BAUD=57 for 57600 baud. Write, reboot and read back. Match the radio serial speed to 57600.
3. Plug the ground radio into the laptop. In Mission Planner SiK Radio settings, read both radios. Match network ID, air rate and radio link settings; save both ends. Keep a supported legal frequency/power configuration.
4. Select the ground radio COM port at 57600 and connect. Tilt the aircraft and verify live attitude, battery readings, GPS and parameter retrieval. A link LED alone is not the full test.
5. Check link statistics and the separate FlySky control link with the installed antennas and all electronics powered. Test telemetry loss separately from RC loss; do not assume they trigger the same failsafe.

Power budget: SiK shares the FC 5 V electronics rail with RX, GPS and airspeed. Include measured radio transmit-current peaks in the 2 A rail test. Keep servo power on Vx/Vx2. The advertised 10 km range is not a demonstrated aircraft range.

Manuals and reference links

Hardware and setup references / checked September 2026

[1] Matek V2 specifications, layout and mapping

<https://www.mateksys.com/?portfolio=f405-wing-v2>

[2] ArduPilot F405-Wing (older details need V2 cross-check)

<https://ardupilot.org/plane/docs/common-matekf405-wing.html>

[3] ArduPilot Choosing Servo Functions

<https://ardupilot.org/plane/docs/servo-functions.html>

[4] ArduPilot board hardware definition (master)

https://github.com/ArduPilot/ardupilot/blob/master/libraries/AP_HAL_ChibiOS/hwdef/MatekF405-Wing/hwdef.dat

[5] FlySky FS-iA10B specification

<https://www.flysky-cn.com/ia10b-canshu>

[6] Matek ASPD-4525

<https://www.mateksys.com/?portfolio=aspd-4525>

[7] Matek ASPD-DLVR

<https://www.mateksys.com/?portfolio=aspd-dlvr>

[8] ArduPilot RC input mapping and reversal

<https://ardupilot.org/plane/docs/rc-throw-trim.html>

[9] ArduPilot V-tail / inverted V-tail setup

<https://ardupilot.org/plane/docs/guide-vtail-plane.html>

[10] ArduPilot Airspeed Calibration

<https://ardupilot.org/plane/docs/calibrating-an-airspeed-sensor.html>

[11] ArduPilot Airspeed Parameters Setup

<https://ardupilot.org/plane/docs/airspeed-parameters-setup.html>

[12] ArduPilot ESC Calibration

<https://ardupilot.org/plane/docs/common-esc-calibration.html>

[13] ArduPilot Plane Failsafe Function

<https://ardupilot.org/plane/docs/apms-failsafe-function.html>

[14] ArduPilot Ground Steering

<https://ardupilot.org/plane/docs/tuning-ground-steering-for-a-plane.html>

[15] Emax ECO II 2807 official drawing in product description

<https://emaxmodel.com/products/emax-eco-ii-series-2807-1300kv-1700kv-1500kv-brushless-motor-for-rc-drone-fpv-racing>

[18] Matek M9N-5883 GNSS and compass

<https://www.mateksys.com/?portfolio=m9n-5883>

[19] TowerPro MG90S specification

<https://towerpro.com.tw/product/mg90s-3/>

[20] ArduPilot SiK telemetry radio

<https://ardupilot.org/plane/docs/common-sik-telemetry-radio.html>

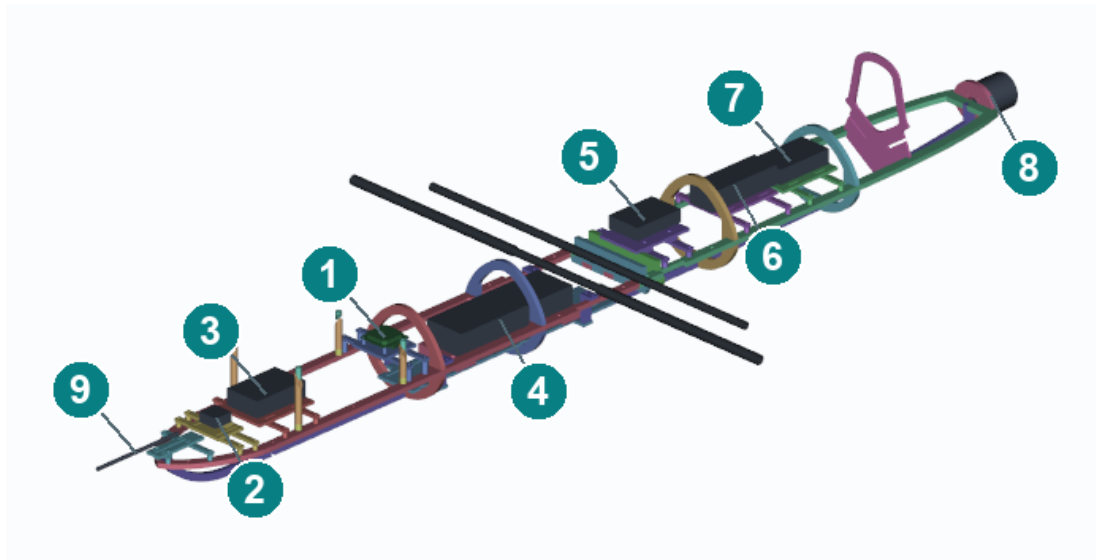
[21] ArduPilot automatic takeoff and hand launch

<https://ardupilot.org/plane/docs/automatic-takeoff.html>

Save the parameter file and connector diagram for the actual aircraft. Complete the loaded-power, sensor, radio and propulsion tests in this guide.

Installed equipment map

Current mounts and access / dimensions are not paper cutting templates



Mark	Placement and attachment
1 / 2 / 9	GPS inside the nose on I09 at Y=-153, antenna up and arrow forward; ASPD-4525 on I13; pitot on I14, aimed forward with pressure ports clear of the nose.
3 / receiver	FS-iA10B on I10 at Y=-300, with independent support for both active antenna tips.
4 / battery	GenX pack on the low central I01 tray. Centre travel Y=-60..+20, or 379.65..459.65 mm aft of the nose. Two straps plus corner stops; battery main and balance leads exit aft.
5 / FC	F405-WING V2 on I04 at Y170, with supplied grommets and 6 mm stand-offs. Arrow/orientation must match setup.
6 / 7	ESC at Y280 on I12; SiK air radio at Y352 on I11. Service through the removable rear foam cover.
8 / motor	Four M3 screws through A26 on the 19 mm bolt circle. Measure engagement and secure the phase leads clear of the bell.

The Manufacturing Team chapter gives fasteners, cover removal and assembly order. The FC is inside the centre fairing; the GPS and airspeed sensor are inside the nose; only the pitot probe projects outside.

Harness routes and balance

Make the installed harness before closing the body

Run	Construction and check
Battery -> FC -> ESC	Use the FC battery/distribution pads, correct polarity and insulated connectors. Start with 12 AWG flexible silicone power wire where extensions are required, matching connector/ESC ratings. Keep DC runs short. Battery leads exit aft; near the GPS bay keep every high-current run at Y=-37 or farther aft, at least 100 mm from the GPS module.
ESC -> motor	Route three phase leads aft along one side of A43. Support extensions and keep them clear of the motor bell, hinges and telemetry antenna. Do not route them through the battery bay.
Servos / wings	Use quality 22-24 AWG servo extensions as required. Label left/right disconnects; support connectors so wing removal cannot pull FC pins. Check loaded voltage at the farthest servo.
GPS / airspeed	Route on the opposite side from high-current wiring. TX/RX cross to their UART partners; SDA/SCL connect in parallel to I2C2. Include ground; insulate and strain-relieve every branch.
Pitot tubing	Keep both tubes distinguishable, unkinked and serviceable. Check the marked total/static ports. Do not blow hard into the sensor or seal a pressure port for calibration.
Antennas	Receiver active tips at right angles, clear of carbon and power leads. Support the 433 MHz antenna separately. Keep GPS/compass clear of ferrous fittings and high-current wiring.

Dry-route wires, add a modest service loop, then measure and cut. Record actual lengths. Extended I2C wiring must pass repeated cold-start detection and a complete motor-on interference check; do not treat a powered LED as a working sensor.

Set balance from measured masses

Replace estimates in Mass_Balance_Worksheet.csv with weighed values. The 522.3 mm aft-nose reference is geometric quarter-MAC, not a flight-approved limit. The sensitivity estimate is about 2.6-3.1 kg, depending on print mass and stated allowances; it is not the aircraft weight.

Balance the complete aircraft, including propeller, connectors and finish. Move the 512 g battery within its 80 mm range, reset both end stops and straps, and recheck lateral balance. If the required location is outside the tray, resolve the weight distribution before flight; do not add unreviewed ballast.

Hand launch and grass landing

Launch method remains unqualified: read Weight and flight assessment

This is a pusher. Prove that motor start occurs only after the aircraft has cleared the launching hand. ArduPilot warns that default takeoff settings can start the motor immediately. [21]

Takeoff control	Review with the installed firmware
TKOFF_THR_MINACC	Launch acceleration trigger. The ArduPilot example is about 15 m/s ² . Check normal handling cannot trigger it and an actual launch motion can.
TKOFF_THR_DELAY	Units are tenths of a second. ArduPilot specifies at least 2, meaning 0.2 s, for pusher hand launches; more may be needed. This is not a guarantee of hand clearance.
TKOFF_THR_MINSPD	Additional GPS-speed gate. The documented starting example is 4 m/s. Check sensor delay and reliability; a gate that never opens is also a failed launch.
TKOFF_THR_IDLE	Keep zero for this stopped-prop launch plan, if the parameter is present. Verify actual motor output at arming and before launch.
Pitch, throttle and modes	Choose TAKEOFF/AUTO behavior, climb limits, abort/recovery and pilot takeover for the measured aircraft. Do not copy generic climb gains or a CG limit as an approved setup.

1. With the motor electrically disconnected and propeller removed, test mode selection, arming, throttle suppression, launch triggering, delay and cancellation. Log the response; repeat after firmware or parameter changes.
2. Before flight, an experienced pilot reviews mass, balance, control authority, thrust, range, failsafes and the launch grip. Use the reinforced fuselage near the measured CG; do not grip a control, foam-only edge, sensor or antenna.
3. Practise the agreed handoff and abort sequence. Use a suitable open area and launch into the wind with wings level. A heavy aircraft must not be treated as a small foam glider merely because it fits in one hand.
4. For landing, use short, even grass and the reviewed power-off approach. Do not rely on a stopped three-blade propeller to eliminate a strike. Check the gear, firewall, propeller and sensor mount after every hard arrival.

Flight-time planning

Estimate from battery capacity / replace assumed current with flight logs

Planning estimate: about 12-20 minutes. This assumes roughly 12-20 A average total current and landing with 20% of the nominal battery capacity unused. Cruise current has not been measured.

Average battery current	Time using 4.16 Ah
8 A	31 minutes
12 A	21 minutes
16 A	16 minutes
20 A	12.5 minutes
25 A	10 minutes
40 A	6 minutes

Calculation

Usable capacity for planning = $5.2 \text{ Ah} \times 0.80 = 4.16 \text{ Ah}$.

Time in minutes = $60 \times 4.16 / \text{average battery current in amperes}$.

Average current must include launch, climb, cruise, servos and avionics. The 40 A ESC rating is not an estimate of cruise current. Static partial-throttle current is not a substitute for a flight log.

Set the first-flight limit

Start with a short evaluation flight and conservative landing reserve. Compare the calibrated current/capacity log with the charger return, voltage under load and the pack condition. Increase duration only after repeated results support it.

The 12-20 minute range is a planning assumption, not an aerodynamic simulation or a demonstrated endurance result. Actual mass, propeller efficiency, speed, wind, manoeuvring and usable battery capacity can move it outside that range.

Parts and materials

Quantities are for one aircraft unless marked as ground equipment. The process coupons and forming tools are not installed parts. Use the CSV for file links and detailed fastening notes.

Plywood

ID	Part	Qty	Material / specification
A01	Rear joiner support	2	5.000 mm aircraft plywood
A04	Aft fuselage former	1	5.000 mm aircraft plywood
A05	Front wing spar	2	5.000 mm aircraft plywood
A07	Rear wing spar	2	5.000 mm aircraft plywood
A10	Outer aileron end plate	2	3.000 mm aircraft plywood
A13	Forward longitudinal former	1	5.000 mm aircraft plywood
A16	Middle fuselage former	1	5.000 mm aircraft plywood
A18	Tail bulkhead	1	2.500 mm aircraft plywood
A24	Middle wing rib	2	5.000 mm aircraft plywood
A25	Front gear bulkhead	1	5.000 mm aircraft plywood
A26	Motor mount	1	5.000 mm aircraft plywood
A34	Forward fuselage former	1	5.000 mm aircraft plywood
A35	Inner wing rib	2	5.000 mm aircraft plywood
A43	Aft longitudinal former	1	5.000 mm aircraft plywood
A44	Outer wing rib	2	4.990 mm aircraft plywood
A47	Main gear base	1	5.000 mm aircraft plywood
A48	Rear gear bulkhead	1	5.000 mm aircraft plywood
A53	Inner aileron end plate	2	3.000 mm aircraft plywood
A54	Central forward former	1	5.000 mm aircraft plywood
A62F	Front nose spine segment	1	5.000 mm aircraft plywood
A62R	Rear spine segment	1	5.000 mm aircraft plywood
A63	Nose cheek doublers (A63/A64)	2	2.500 mm aircraft plywood
G03	Main gear mounting doubler	1	3.000 mm aircraft plywood
H05B	Control horn backing plate	4	2.500 mm aircraft plywood
I01	Adjustable battery tray	1	3.000 mm aircraft plywood

Printed parts

ID	Part	Qty	Material / specification
A11	Left aileron	1	LW-PLA preferred; PLA+ mass must be budgeted
A21	Right aileron	1	LW-PLA preferred; PLA+ mass must be budgeted
A23	Left aileron hinge	1	LW-PLA preferred; PLA+ mass must be budgeted
A36	Right aileron hinge	1	LW-PLA preferred; PLA+ mass must be budgeted
A45	Right wing root	1	PLA+ or PETG; functional coupon required
A55	Left wing root	1	PLA+ or PETG; functional coupon required
I04	FC Bridge	1	qualified PETG / PLA+

ID	Part	Qty	Material / specification
I05-1	Nose Cover Column	1	qualified PETG / PLA+
I05-2	Nose Cover Column	1	qualified PETG / PLA+
I05-3	Nose Cover Column	1	qualified PETG / PLA+
I05-4	Nose Cover Column	1	qualified PETG / PLA+
I07	FC Service Hatch	1	original shell print material
I08-1	Hatch Support Tab	1	qualified PETG / PLA+
I08-2	Hatch Support Tab	1	qualified PETG / PLA+
I13	Airspeed Board Carrier	1	PETG / PLA+
A37	Nose gear mount	1	qualified PLA+ or PETG
A09	Nose steering bearing	1	qualified PLA+ or PETG
I02	Battery Corner Stop	2	qualified PETG / PLA+
I17-1L	Battery Hanger	1	qualified PETG / PLA+
I17-1R	Battery Hanger	1	qualified PETG / PLA+
I17-2L	Battery Hanger	1	qualified PETG / PLA+
I17-2R	Battery Hanger	1	qualified PETG / PLA+
I00	Nose Cover Spacer 6mm	4	nylon or qualified print material
I10	Nose Receiver Carrier	1	qualified PETG / PLA+
A06	Centre fairing	1	original shell print material
I18	Battery Access Hatch	1	original shell print material
I19-1	Battery Hatch Tab	1	qualified PETG / PLA+
I19-2	Battery Hatch Tab	1	qualified PETG / PLA+
I12	Forward ESC Carrier	1	qualified PETG / PLA+
I11	Aft Telemetry Carrier	1	qualified PETG / PLA+
A12	Tail fairing	1	original shell print material
A59	Nose shell	1	PLA+ / nylon fittings
I09	Internal GPS Bridge	1	PLA+ / nylon fittings
I14	Slim Pitot Carrier	1	PLA+ / nonmagnetic fittings
I06-1	Compact Cover Washer	1	PLA+ / nonmagnetic fittings
I06-2	Compact Cover Washer	1	PLA+ / nonmagnetic fittings
I06-3	Compact Cover Washer	1	PLA+ / nonmagnetic fittings
I06-4	Compact Cover Washer	1	PLA+ / nonmagnetic fittings

Foam patterns

ID	Part	Qty	Material / specification
F01	Left inner wing	1	5 mm FliteBoard Pro
F02	Right inner wing	1	5 mm FliteBoard Pro
F03	Right outer wing	1	5 mm FliteBoard Pro
F04	Left outer wing	1	5 mm FliteBoard Pro
F05	Right fixed tail - main	1	5 mm FliteBoard Pro
F06	Left fixed tail - main	1	5 mm FliteBoard Pro

ID	Part	Qty	Material / specification
F07	Right fixed tail - root collar	1	5 mm FliteBoard Pro
F08	Left fixed tail - root collar	1	5 mm FliteBoard Pro
F09	Forward lower fuselage - left curve	1	5 mm FliteBoard Pro
F10	Forward lower fuselage - right curve	1	5 mm FliteBoard Pro
F11	Middle lower fuselage - right curve	1	5 mm FliteBoard Pro
F12	Middle lower fuselage - left curve	1	5 mm FliteBoard Pro
F13	Forward lower fuselage - left lip	1	5 mm FliteBoard Pro
F14	Forward lower fuselage - bottom strip	1	5 mm FliteBoard Pro
F15	Forward lower fuselage - right lip	1	5 mm FliteBoard Pro
F16	Middle lower fuselage - left lip	1	5 mm FliteBoard Pro
F17	Middle lower fuselage - right lip	1	5 mm FliteBoard Pro
F18	Middle lower fuselage - bottom strip	1	5 mm FliteBoard Pro
F19	Aft lower fuselage - left lip	1	5 mm FliteBoard Pro
F20	Aft lower fuselage - left curve	1	5 mm FliteBoard Pro
F21	Aft lower fuselage - right curve	1	5 mm FliteBoard Pro
F22	Aft lower fuselage - right lip	1	5 mm FliteBoard Pro
F23	Aft upper fuselage - left	1	5 mm FliteBoard Pro
F24	Aft upper fuselage - right	1	5 mm FliteBoard Pro
F25	Forward upper fuselage - right	1	5 mm FliteBoard Pro
F26	Forward upper fuselage - left	1	5 mm FliteBoard Pro
F27	Rear underside band 1 - forward	1	5 mm FliteBoard Pro
F28	Rear underside band 2	1	5 mm FliteBoard Pro
F29	Rear underside band 3	1	5 mm FliteBoard Pro
F30	Rear underside band 4 - aft	1	5 mm FliteBoard Pro
F31	Right tail control laminate blank	2	5 mm FliteBoard Pro
F32	Left tail control laminate blank	2	5 mm FliteBoard Pro
F33	Aft lower fuselage - bottom strip	1	5 mm FliteBoard Pro

Carbon tubes

ID	Part	Qty	Material / specification
T01	T01 front joiner	1	Carbon OD8/ID6 mm
T02	T02 right front sleeve	1	Carbon OD10/ID8.2 mm
T03	T03 left front sleeve	1	Carbon OD10/ID8.2 mm
T04	T04 rear joiner	1	Carbon OD8/ID6 mm

Hardware

ID	Part	Qty	Material / specification
H05	Control horns	4	Purchased lightweight RC horns
H07	Pushrod and clevis sets	5	Matched RC linkage

ID	Part	Qty	Material / specification
H08	Aileron pins	2	3 mm smooth rod; start 285 mm
G01	Main landing gear	1	20 x4 mm 6061-T6 aluminium
G02	Steerable nose gear	1	4 mm spring wire, shop-qualified forming
W01	Lightweight wheels	3	76.2 mm OD, 22 mm nominal hub width, 4 mm bore
H10	Electronics carrier fasteners	20	M2 x16 starting length, washers and locking nuts
H11	Airspeed carrier fasteners	6	Four M2 x16 foot bolts + two M2 PCB screws
H12	Battery hanger fasteners	8	Four M2 x16 foot bolts + four M2 finished 35-37 mm tray bolts
H13	Battery stop fasteners	4	M2 x12 starting length, 4-5 mm OD washers, locking nuts
H14	Nose cover fasteners	4	Nonmagnetic M2 brass/nylon: two x80 and two x60 starting lengths, washers/locking nuts
H15	Internal GPS fasteners	8	Four M2 x25 nylon bridge bolts + four M2 x14 nylon PCB bolts; washers and nuts
H16	FC stand-offs	4	M3 nylon 6 mm spacers, screws and supplied grommets
H17	Servo mounting fasteners	10	M2 through-bolts, broad washers and locking nuts
H18	Cable and sensor installation	1 set	Soft straps, tie bases, grommets, extension leads, heat-shrink, XPS-safe adhesive
H20	Wing root retainers	6	M2 x20 starting length; 12 washers and 6 locking nuts
H21	Nose bypass fasteners	4	Two M2 x40-45 inner bolts and two M2 x16 outer bolts; washers and locking nuts
H22	Nose bearing fasteners	4	M3 x16-20 starting length; washers and locking nuts
H23	Motor screws	4	M3 with washers; length from measured thread depth
H24	Control horn fasteners	8	M2 through-bolts, washers, locking nuts and crush-resistant spacer sleeves
H25	Pitot tab fastener	1	M2 x10 starting length, two washers and locking nut; two small anti-rotation ties
H01	Main gear bolts	2	M3 x20 starting length, broad washers and locking nuts
H02	Main wheel axles	2	M4 x40 starting length, smooth shank, spacers and locking nuts
H03	Nose arm and collars	1 arm + 4 collars	4 mm clamp steering arm and 4 mm shaft collars
H04	Battery straps	2	20 mm wide, at least 250 mm long, non-stretch hook-and-loop

Electronics

ID	Part	Qty	Material / specification
E01	Flight controller	1	Matek F405-WING V2
E02	GNSS and compass	1	Matek M9N-5883
E03	Airspeed kit	1	Matek ASPD-4525 with pitot and two tubes
E04	Receiver	1	FlySky FS-IA10B
E05	ESC	1	ReadytoSky 40 A, 2-4S, 5 V /3 A BEC

ID	Part	Qty	Material / specification
E06	Flight battery	1	GenX 4S 5200 mAh 40C; 138 x43 x33 mm, 512 g
E07	Motor	1	Emax ECO II 2807 1300 KV
E08	Propeller	1	Gemfan 9045CL, 9-inch three-blade, selected handedness
E09	Telemetry pair	1	SiK 433 MHz, user-specified 100 mW; only air radio flies
E10	Transmitter	1	FlySky FS-i6X; ground equipment
E11	Servos	5	TowerPro MG90S; nominal 13.4 g each

Hardware and forming

Gear and wheels

G01 is 20 x4 mm 6061-T6 aluminium with a 12 mm inside bend radius. The formed STEP governs final dimensions. $K=0.40$ in the flat DXF is a starting assumption: qualify a bend coupon, correct the allowance for the actual tools, then form and finish-drill. Do not treat the ideal bending calculation as a landing rating.

Nose gear

G02 is 4 mm spring wire, formed by a shop able to preserve the required material properties. A09/A37 bores are 4.2 mm before process adjustment. Fit a 4 mm clamp steering arm and positive shaft retainers. Use the 76.2 mm wheel reference, 22 mm nominal hub and 4 mm bore. The nose axle ends at X46; fit washers/collars without clamping the hub.

Main attachment

Bond G03 below A47. Two M3 through-bolts clamp A47/G03/G01 with broad washers and locking nuts. Use smooth-shank M4 main axles. Select actual bolt lengths for the measured stack and retain two threads beyond the locking nut.

Motor

A26 is 5 mm aircraft plywood. Its four 3.2 mm holes lie on a 19 mm bolt circle, with adjacent centres 13.435 mm apart. Preserve the 12 mm centre opening. Measure M3 screw engagement to avoid windings. The 33.9 x34 mm conservative motor envelope clears the tail fairing; verify the actual bell and leads before running.

Controls

Use five TowerPro MG90S, ten mounting bolts, four purchased lightweight horns, four H05B backing plates and five adjustable pushrods with ten retained rod ends. Align horn pins over hinge axes. Through-fasteners and crush-resistant spacer sleeves carry loads across hollow prints or foam. Set neutral geometry before cutting rods; retain pins and clips mechanically.

Wing pins and retainers

Two smooth 3 mm aileron hinge rods start at 285 mm and are trimmed after fitting retainers, leaving about 1 mm axial freedom. Use three M2 root retainers per wing. Bond front sleeves only; do not glue the sliding joiner into both wings.

Battery and cover hardware

Use the quantities in Complete_Parts_and_Materials.csv. Battery hangers attach to A13; the tray hangs below them. Two straps wrap the pack and tray, with two padded corner stops. Fit nonmagnetic cover/GPS hardware near the compass. Front-stop access must be proved with the chosen screws and tools before covering.

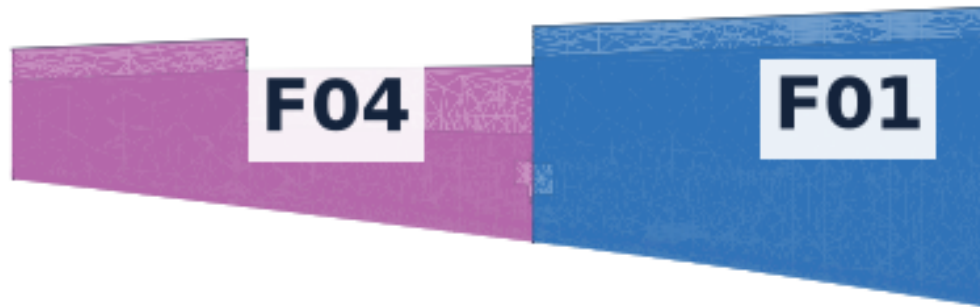
Process checks

Use the plywood slot coupons and 80 x35 x4 mm printer coupon. Current printer-coupon bores are 2.0, 2.2, 3.0, 3.2, 4.0, 4.2, 4.4, 8.0, 8.2 and 10.2 mm. Qualify the actual stock, kerf, printed fit, foam forming and adhesive before making the full set.

Wings: match each skin to its side and span section

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

LEFT AIRCRAFT SIDE (-X)



RIGHT AIRCRAFT SIDE (+X)



 F04 Left outer wing A27

 F01 Left inner wing A52

 F02 Right inner wing A49

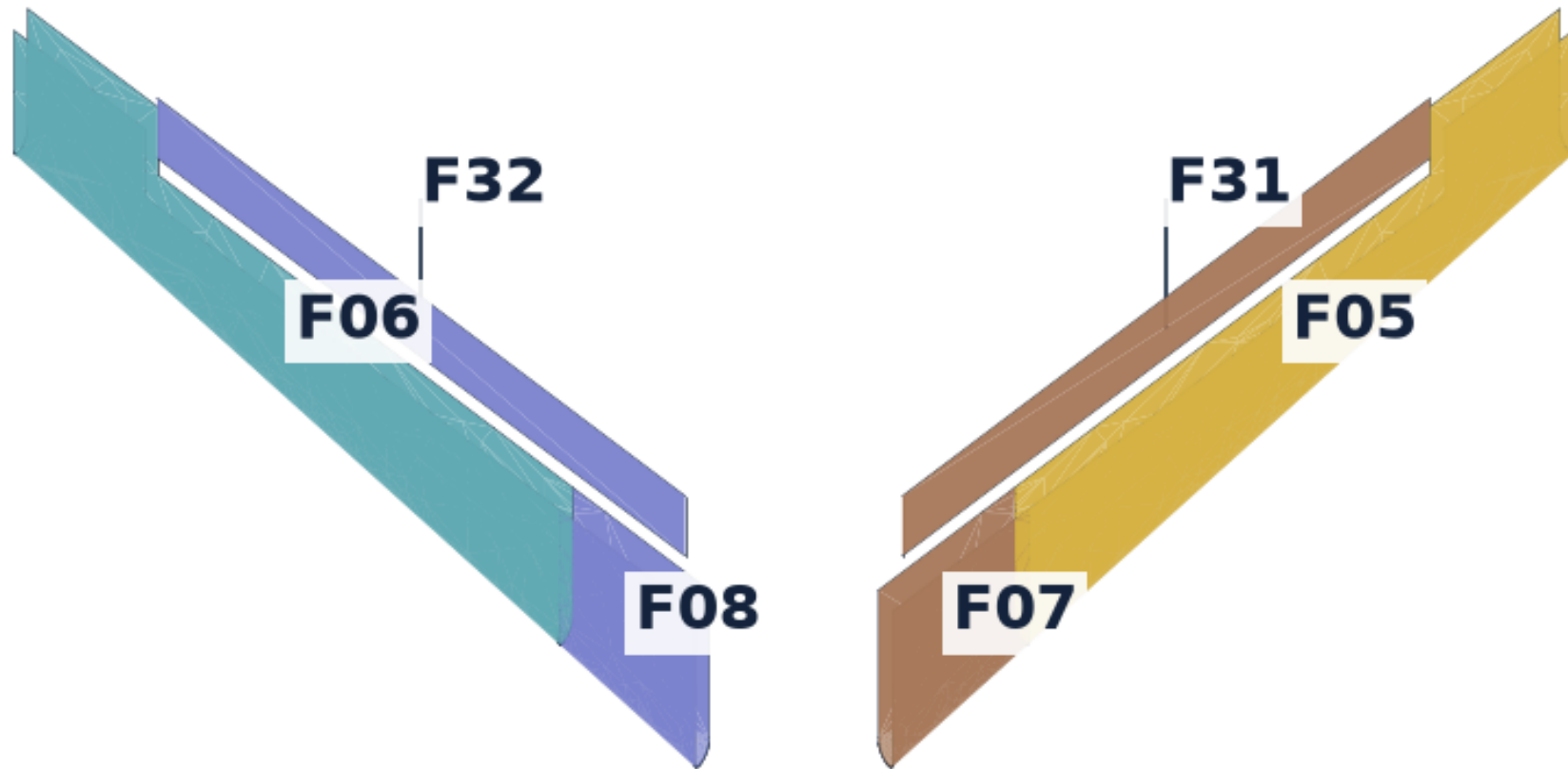
 F03 Right outer wing A38

Fit inner and outer sections to their ribs and spars before closing the trailing seams.

The narrow hinge-rail contact zones need inner-core relief; preserve the outer paper. See the forming guide.

V-tail: two fixed skins and one moving control on each side

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates



- F06 Left main skin A22
- F08 Left root collar A22
- F32 Left control A60: cut two layers

- F05 Right main skin A57
- F07 Right root collar A57
- F31 Right control A41: cut two layers

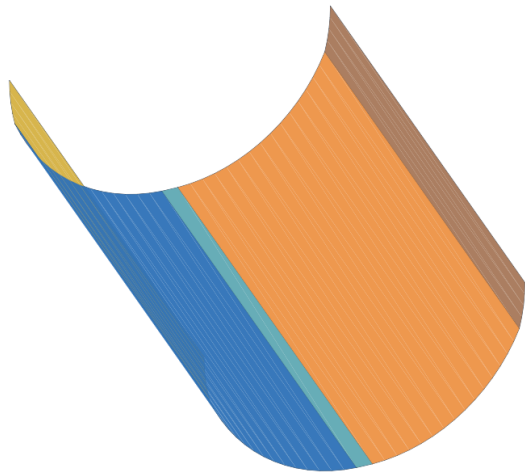
Each fixed-tail pair meets at the root collar seam (80% of the CAD span parameter from the tip).

Control highlights show the CAD broad faces. Laminate two 5 mm blanks to 10 mm, then shape the R5 edges.

Lower fuselage: five distinct pieces around each section

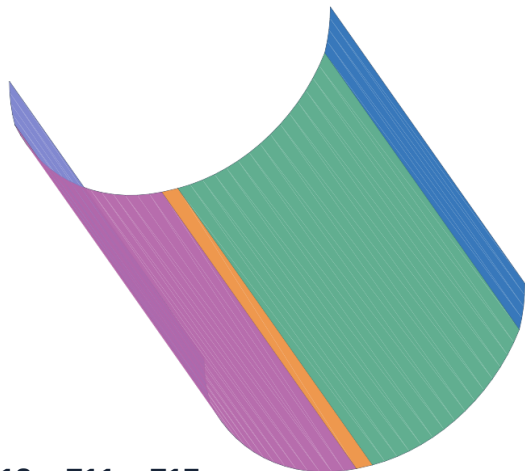
MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

FORWARD LOWER SHELL A56 | Y -335 to -130 mm

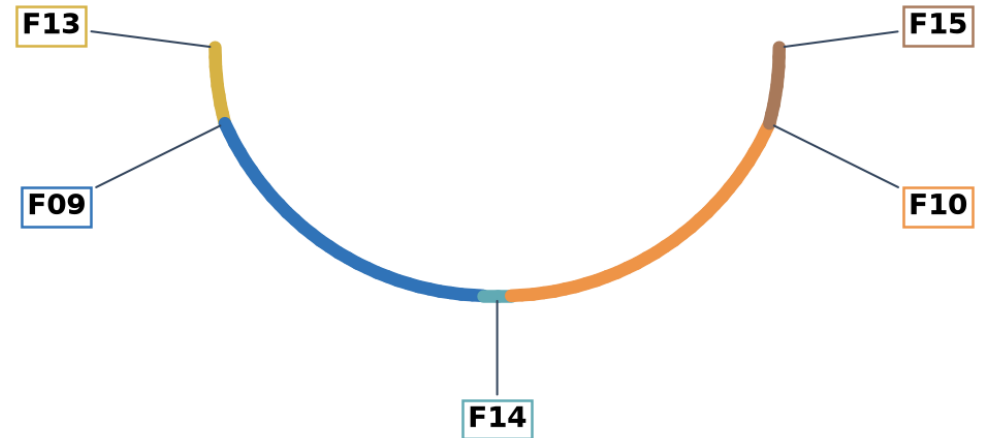


F13 > F09 > F14 > F10 > F15

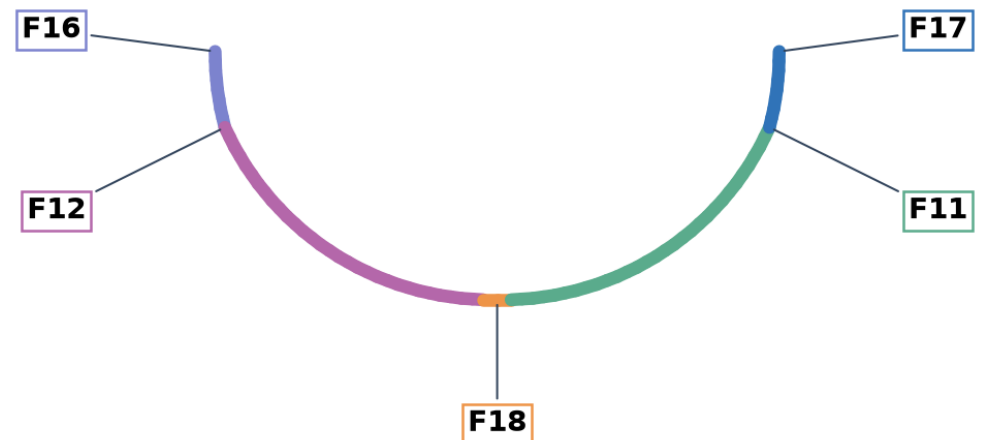
MIDDLE LOWER SHELL A17 | Y -130 to +238.65 mm



F16 > F12 > F18 > F11 > F17



Cross-section: left to right around bottom

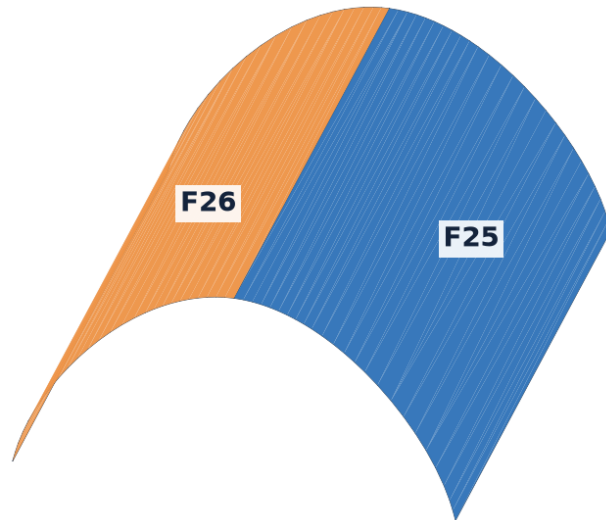


The narrow bottom strips are real parts; retain them.

Upper and aft-lower fuselage: match the coloured face pairs

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

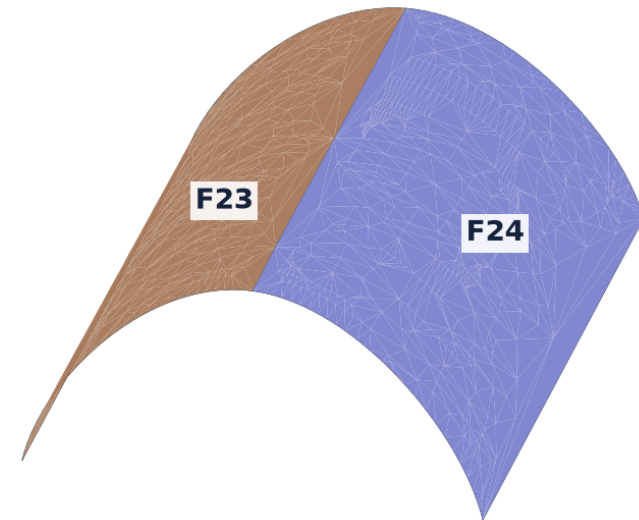
FORWARD UPPER A08 | Y -130 to 0



F26 left / F25 right; meet at the top.

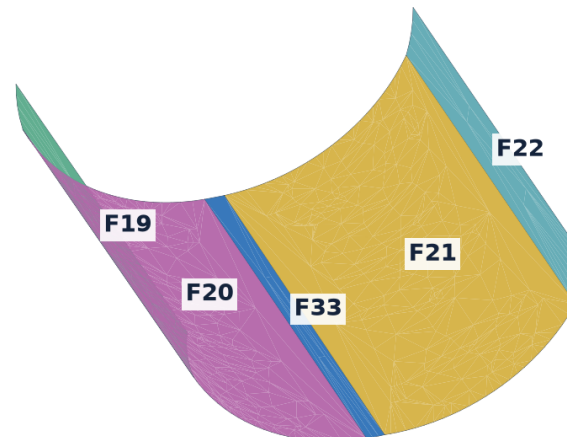
F23/F24 include a 1 mm paper-only lower-edge return: remove its foam, then trim the paper to fit.

AFT UPPER A15 | Y 233.65 to 389.65



F23 left / F24 right; meet at the top.

AFT LOWER A42 | Y 238.65 to 389.65 mm



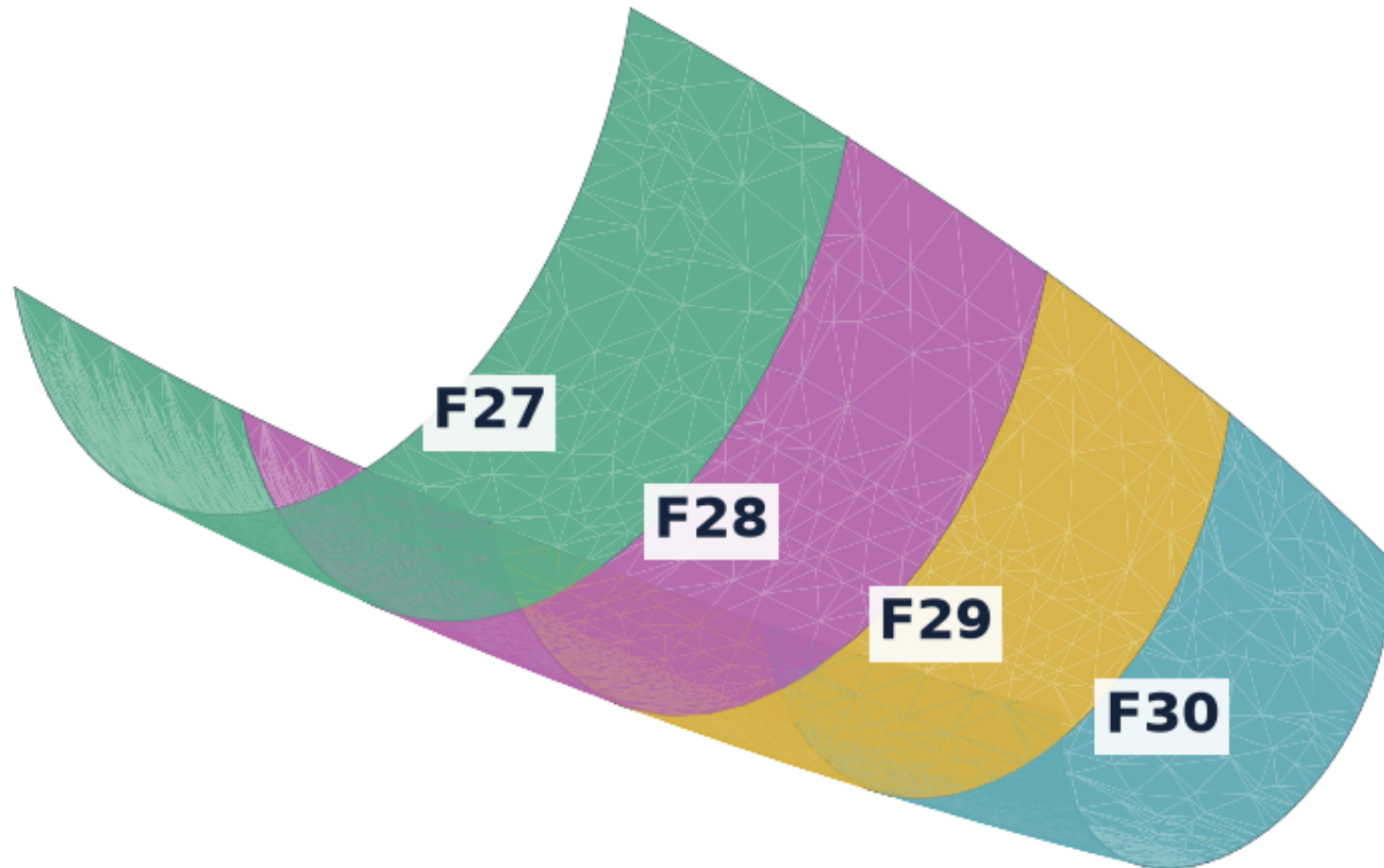
Around the underside: F19 left lip > F20 left curve > F33 bottom > F21 right curve > F22 right lip.

Fit the printed transitions between these shell groups before final skin bonding.

Use the numbered 1:1 DXFs / tiled PDF for cutting. Preserve the outside paper. See the forming guide.

Rear underside: assemble four bands from front to aft

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates



■ F27 1 - forward band

■ F28 2

■ F29 3

■ F30 4 - aft band

A20 assembly order: F27 > F28 > F29 > F30. The Y coordinate increases toward the tail.

Butt the outside-paper seam lines. Add separate inside straps; dry-fit around A43 and A62 before gluing.

Foam / form and fit

Stock

5 mm FliteBoard Pro paper-faced XPS, 1000 x600 mm. Measure thickness. Keep the outside paper as the developed reference; do not resize patterns for different stock.

Cut

Print at 100%, measure the 100 x20 mm rectangle and align matching registration marks. Knife-cut the CUT line. Mark the part ID and OUTSIDE before peeling.

Form

Test the tightest curve on scrap. Peel inside paper only where needed. Relieve/compress the core gradually over the real frame; do not stretch a short edge into place.

Seams

Bevel the inside core to the actual meeting angle. Butt outside-paper edges and add separate 15 mm internal paper straps. Keep adhesive and straps out of hinges. Hold alignment through cure.

Allowances

F23/F24 have a 1 mm paper-only lower lip. Other skins have no added lap/trim. F31/F32 are two-layer moving controls with 5 mm sacrificial stock before shaping. Use an XPS-compatible adhesive.

Foam / assembly sequence

Wings

Fit from the leading-edge region toward the open seam. Relieve only inner foam for ribs and hinge rails. On F01/F02, transfer A44 rib contacts to the inner core and relieve until dry-fit contact is removed; keep outside paper intact. Dry-fit a complete wing and moving surface before bonding. Check incidence, symmetry and twist.

Fuselage

Use the panel map. Fit lower forward and middle panels, then the rear underside. A42 underside order is F19, F20, F33, F21, F22. F27-F30 are the numbered rear-underside bands. Support every inner butt seam. Relieve only inside core in F19-F22/F33 and F27-F30 where A43 or A62R contacts it. Transfer these contacts from the assembled frame; do not cut the wood to force a skin fit.

Fixed tails

Use removable split collars on a straight jig: T03 at 6.84 mm, T02 at 171 mm, T01 at 335.16 mm along the 342 mm span. Remove both collar halves after cure. Keep the formed shape; do not add ribs into the filled section.

Moving tails

Laminate two 5 mm blanks per control. Trim to the finished outline and shape long edges to R5. Tape-hinge on the indicated axis, then fit backed horns and linkages. Verify combined travel.

Checks

Preserve outside paper around local reliefs. Reinforce seams without recreating a hard contact. Weigh completed skins and inspect them for warping or damage.

Foam / removable covers and ports

F25/F26

Join and form the two roof skins first. Make a service lid on the formed roof: X +/-25, Y=-115..-15. These are plan-view assembly coordinates, not a flat cutting pattern. Add 15 mm internal paper supports and replaceable reinforced-tape retention.

F23/F24

Join into one formed cover. Retain its perimeter on internal paper supports with six reinforced-tape tabs. Keep it removable for ESC/radio access. Do not permanently glue its lower lips to the body.

Cooling

On the formed rear cover, make a 24 x12 mm inlet at Y250 and a 30 x20 mm outlet at Y374. Add a small forward-facing scrap-foam scoop over the inlet, directing air toward the ESC. Confirm temperature in the loaded test.

Gear

Transfer actual installed gear crossings after forming. Give main legs 3 mm nominal clearance and the vertical nose shaft a 6 mm opening. Reinforce the paper border. Check steering and loaded deflection; skin must not carry landing loads.

Before closure

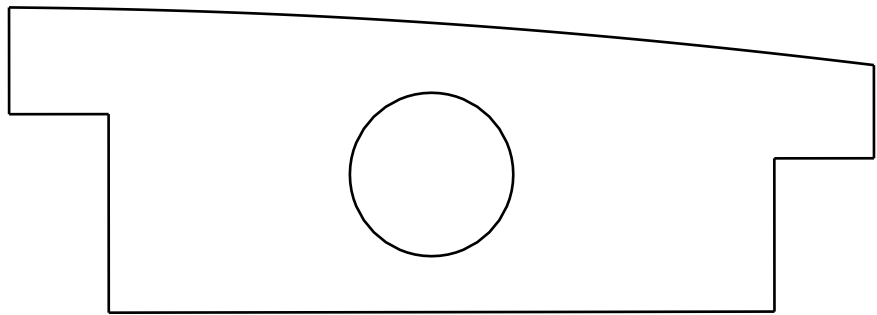
Prove battery insertion/removal, strap and stop access, sensor tubing, connectors, antenna support and full control travel. Leave pull strings and label all disconnects.

Plywood / 25 profiles / 37 pieces

Preview only. Use the millimetre DXFs or actual-size templates. Blue sanding references are not cuts.

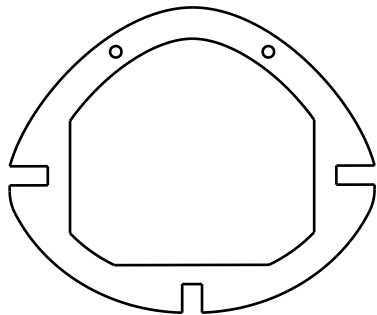
A01 / Rear joiner support

5.000 mm / quantity 2



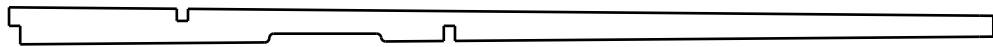
A04 / Aft fuselage former

5.000 mm / quantity 1



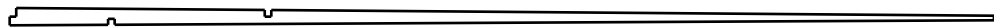
A05 / Front wing spar

5.000 mm / quantity 2



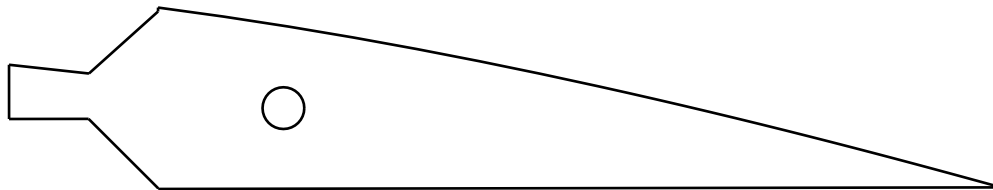
A07 / Rear wing spar

5.000 mm / quantity 2



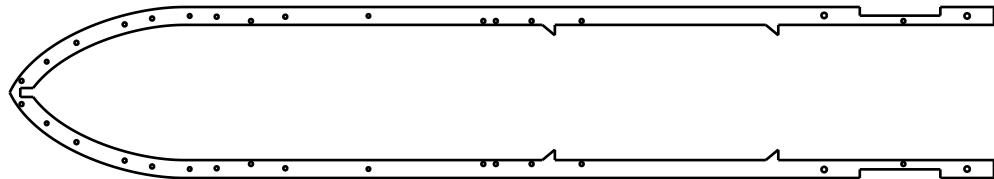
A10 / Outer aileron end plate

3.000 mm / quantity 2



A13 / Forward longitudinal former

5.000 mm / quantity 1

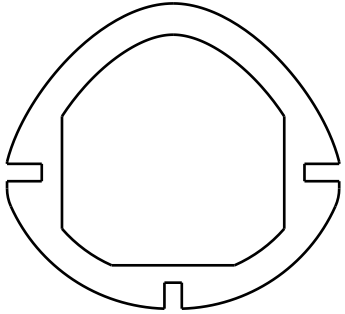


Plywood / 25 profiles / 37 pieces

Preview only. Use the millimetre DXFs or actual-size templates. Blue sanding references are not cuts.

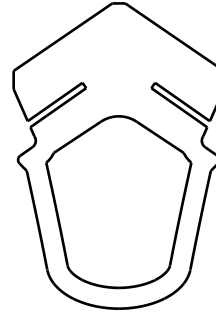
A16 / Middle fuselage former

5.000 mm / quantity 1



A18 / Tail bulkhead

2.500 mm / quantity 1



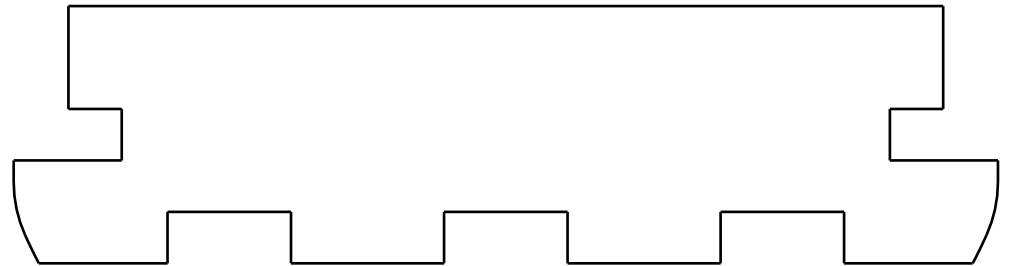
A24 / Middle wing rib

5.000 mm / quantity 2



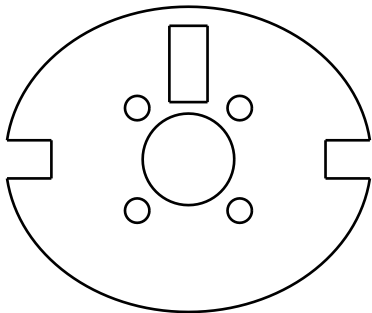
A25 / Front gear bulkhead

5.000 mm / quantity 1



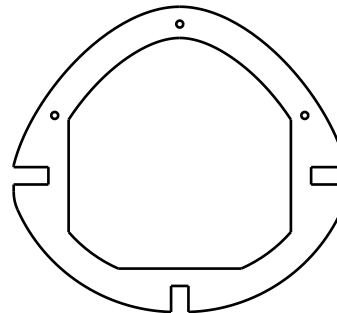
A26 / Motor mount

5.000 mm / quantity 1



A34 / Forward fuselage former

5.000 mm / quantity 1

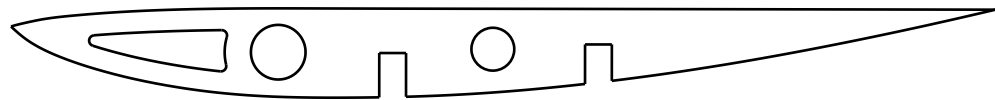


Plywood / 25 profiles / 37 pieces

Preview only. Use the millimetre DXFs or actual-size templates. Blue sanding references are not cuts.

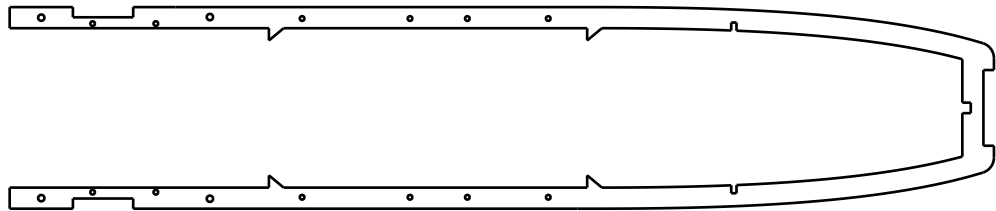
A35 / Inner wing rib

5.000 mm / quantity 2



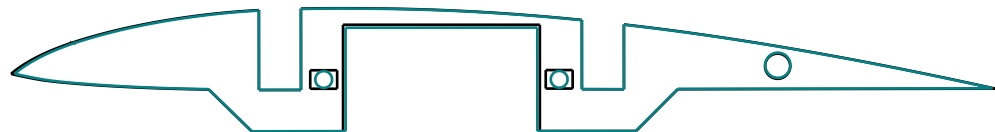
A43 / Aft longitudinal former

5.000 mm / quantity 1



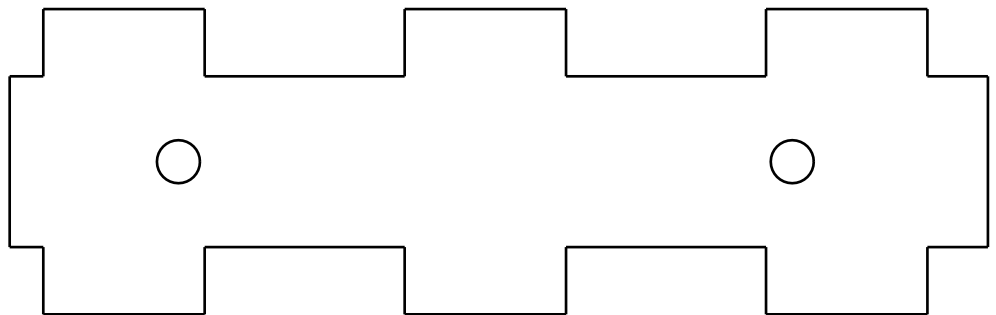
A44 / Outer wing rib

4.990 mm / quantity 2



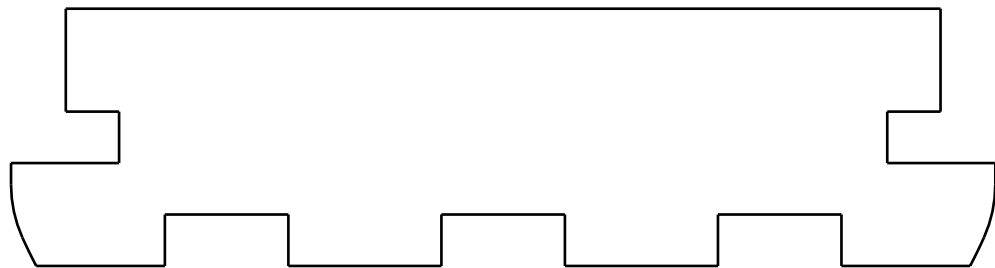
A47 / Main gear base

5.000 mm / quantity 1



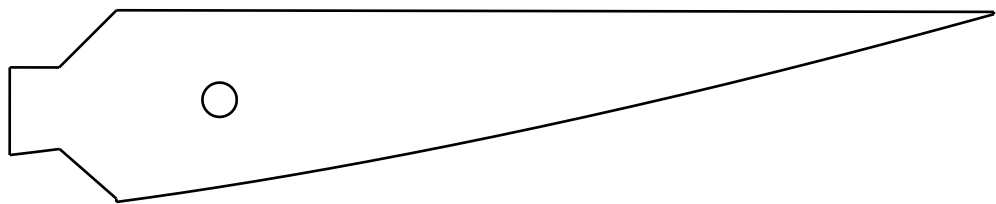
A48 / Rear gear bulkhead

5.000 mm / quantity 1



A53 / Inner aileron end plate

3.000 mm / quantity 2

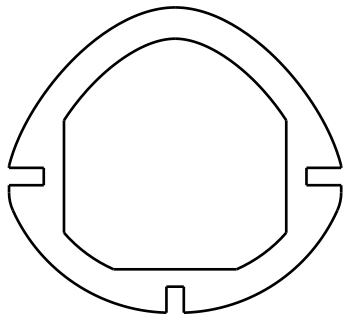


Plywood / 25 profiles / 37 pieces

Preview only. Use the millimetre DXFs or actual-size templates. Blue sanding references are not cuts.

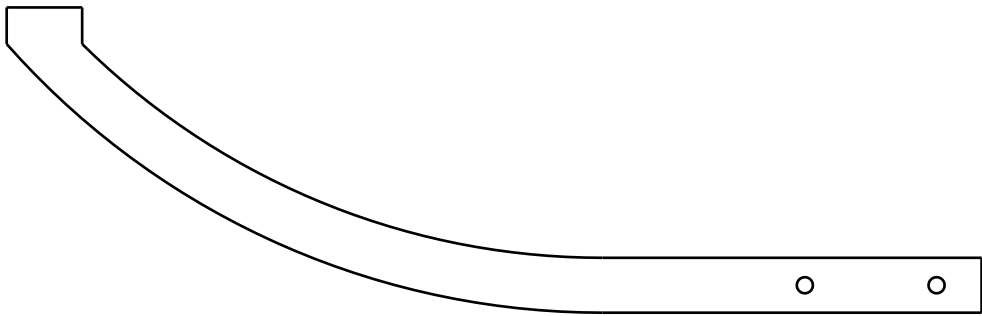
A54 / Central forward former

5.000 mm / quantity 1



A62F / Front nose spine segment

5.000 mm / quantity 1



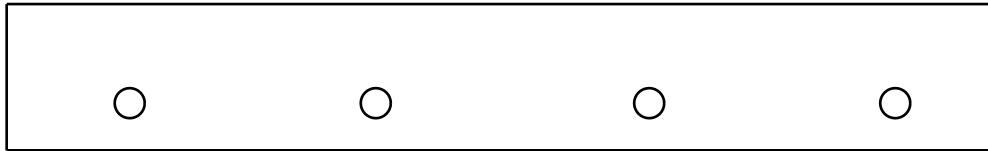
A62R / Rear spine segment

5.000 mm / quantity 1



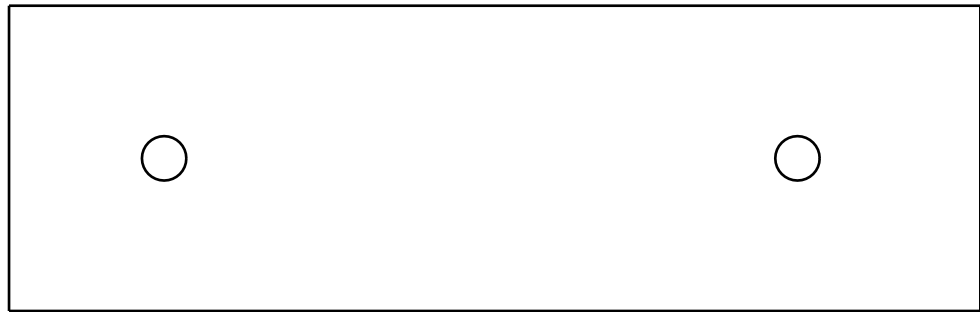
A63 / Nose cheek doublers (A63/A64)

2.500 mm / quantity 2



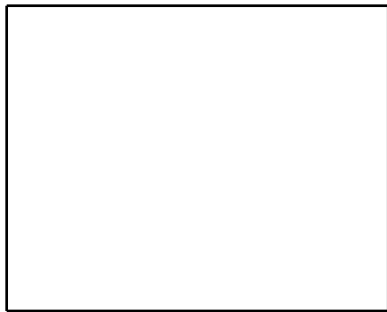
G03 / Main gear mounting doubler

3.000 mm / quantity 1



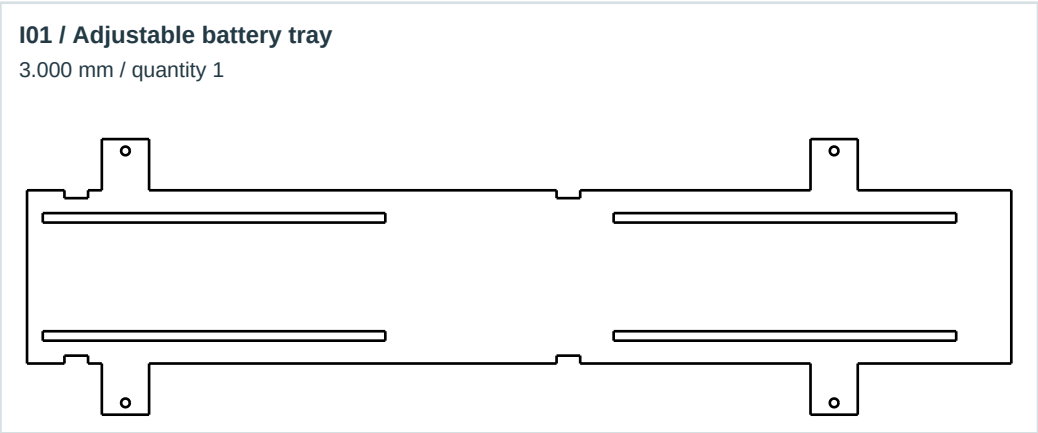
H05B / Control horn backing plate

2.500 mm / quantity 4



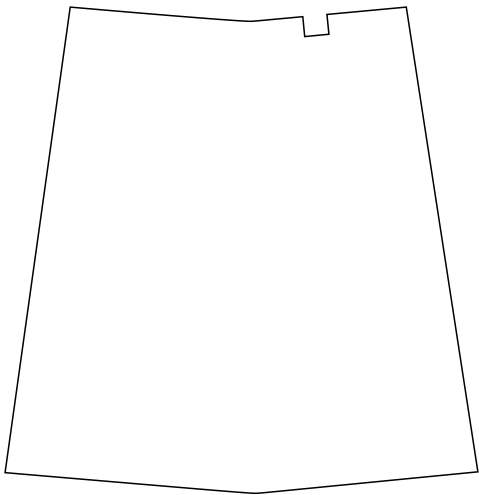
Plywood / 25 profiles / 37 pieces

Preview only. Use the millimetre DXFs or actual-size templates. Blue sanding references are not cuts.

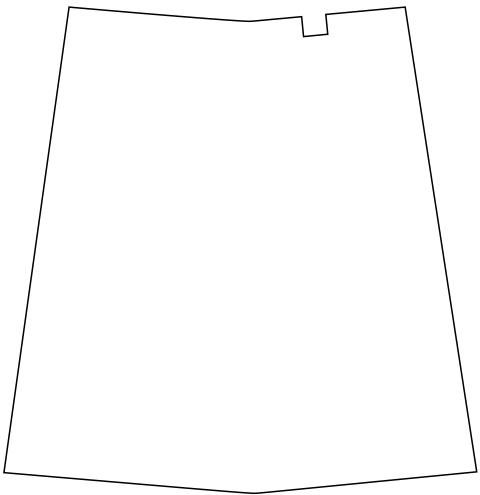


MQ9 | developed foam patterns | overview only

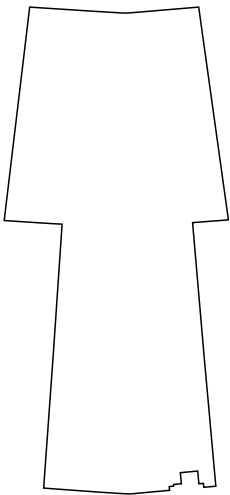
Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



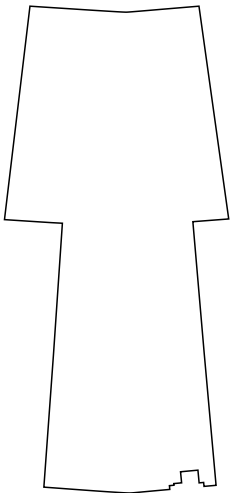
F01 Developed skin
wing_inner_A
443.1 x 455.6 mm | max strain 0.341%



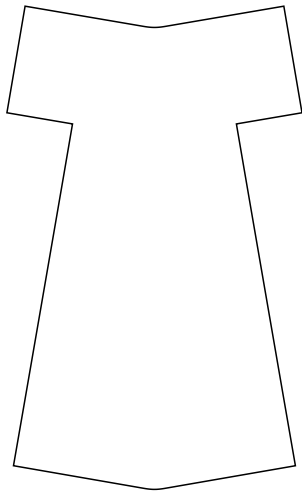
F02 Developed skin
wing_inner_B
443.1 x 455.6 mm | max strain 0.341%



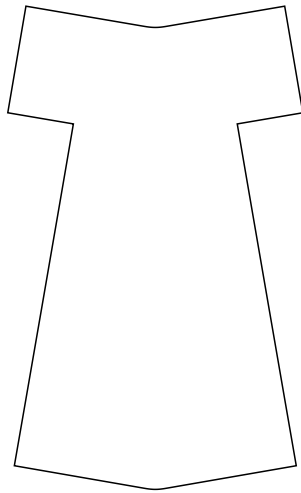
F03 Developed skin
wing_outer_A
237.1 x 514.7 mm | max strain 0.357%



F04 Developed skin
wing_outer_B
237.1 x 514.7 mm | max strain 0.366%



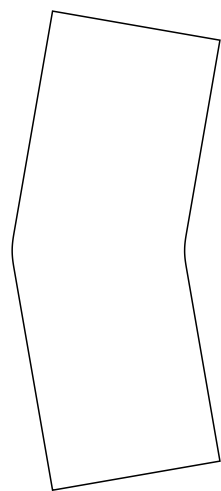
F05 Developed skin
tail_A_rootcollar1
177.4 x 290.5 mm | max strain 0.278%



F06 Developed skin
tail_B_rootcollar1
177.4 x 290.5 mm | max strain 0.278%

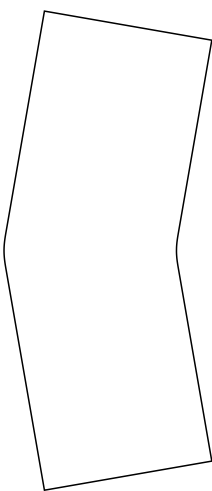
MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F07 Developed skin

tail_A_rootcollar2
83.6 x 192.9 mm | max strain 0.174%



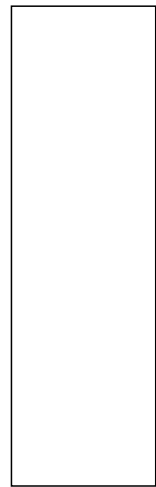
F08 Developed skin

tail_B_rootcollar2
83.6 x 192.9 mm | max strain 0.174%



F09 Developed skin

fuselage_forward_A
61.6 x 205.0 mm | max strain 0.023%



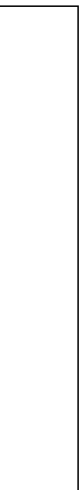
F10 Developed skin

fuselage_forward_B
61.6 x 205.0 mm | max strain 0.023%



F11 Developed skin

fuselage_middle_A
61.6 x 368.7 mm | max strain 0.023%



F12 Developed skin

fuselage_middle_B
61.6 x 368.6 mm | max strain 0.023%

MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F13 Developed skin

fuselage_src3_face8
14.4 x 205.0 mm | max strain 0.018%



F14 Developed skin

fuselage_src3_face10
5.3 x 205.0 mm | max strain 0.000%



F15 Developed skin

fuselage_src3_face12
14.4 x 205.0 mm | max strain 0.019%



F16 Developed skin

fuselage_src21_face2
14.4 x 368.7 mm | max strain 0.018%



F17 Developed skin

fuselage_src21_face16
14.4 x 368.7 mm | max strain 0.019%



F18 Developed skin

fuselage_src21_face18
5.3 x 368.7 mm | max strain 0.000%

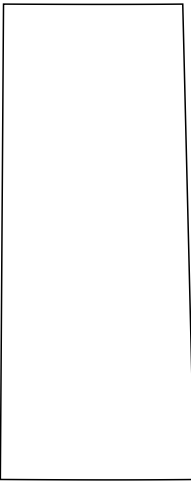
MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



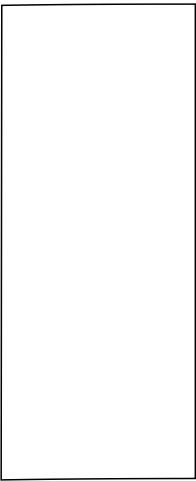
F19 Developed skin

fuselage_src10_face16
14.6 x 151.0 mm | max strain 0.000%



F20 Developed skin

fuselage_src10_face17
61.6 x 151.1 mm | max strain 0.037%



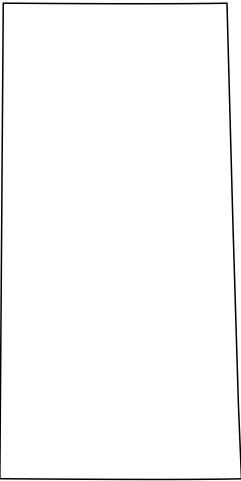
F21 Developed skin

fuselage_src10_face19
61.8 x 151.4 mm | max strain 0.035%



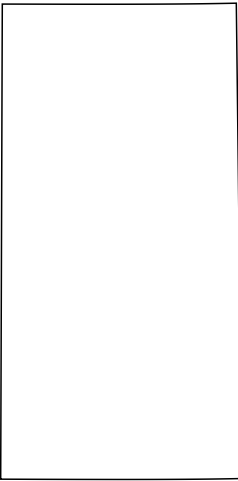
F22 Developed skin

fuselage_src10_face20
14.8 x 151.0 mm | max strain 0.000%



F23 Developed skin

fuselage_src22_face11
79.3 x 156.2 mm | max strain 0.030%

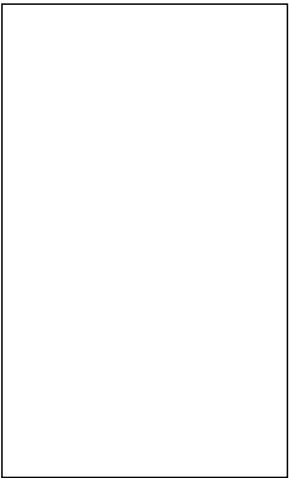


F24 Developed skin

fuselage_src22_face13
78.8 x 156.4 mm | max strain 0.037%

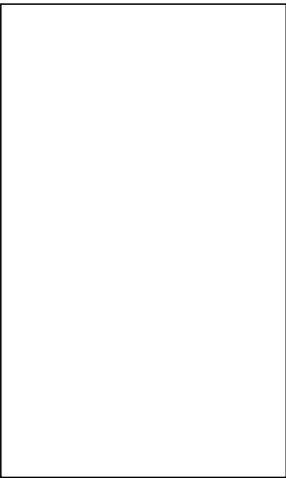
MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



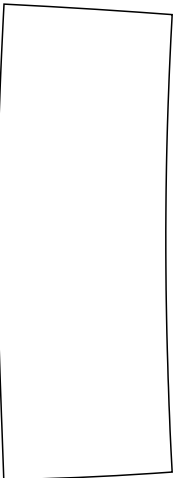
F25 Developed skin

fuselage_src25_face6
78.4 x 130.0 mm | max strain 0.000%



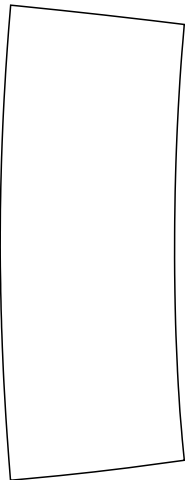
F26 Developed skin

fuselage_src25_face8
78.4 x 130.0 mm | max strain 0.000%



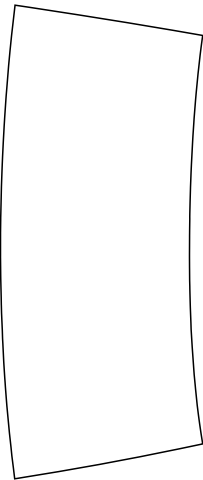
F27 Developed skin

fuselage_src18_face14_axis1_strip1of4
55.2 x 150.2 mm | max strain 0.287%



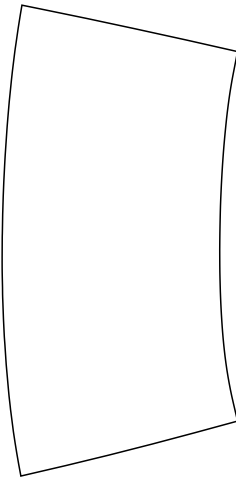
F28 Developed skin

fuselage_src18_face14_axis1_strip2of4
55.9 x 144.4 mm | max strain 0.295%



F29 Developed skin

fuselage_src18_face14_axis1_strip3of4
56.5 x 132.3 mm | max strain 0.336%



F30 Developed skin

fuselage_src18_face14_axis1_strip4of4
57.0 x 113.9 mm | max strain 0.321%

MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F31 Control blank

tail_control_blank_A
45.0 x 275.0 mm | max strain 0.000%



F32 Control blank

tail_control_blank_B
45.0 x 275.0 mm | max strain 0.000%

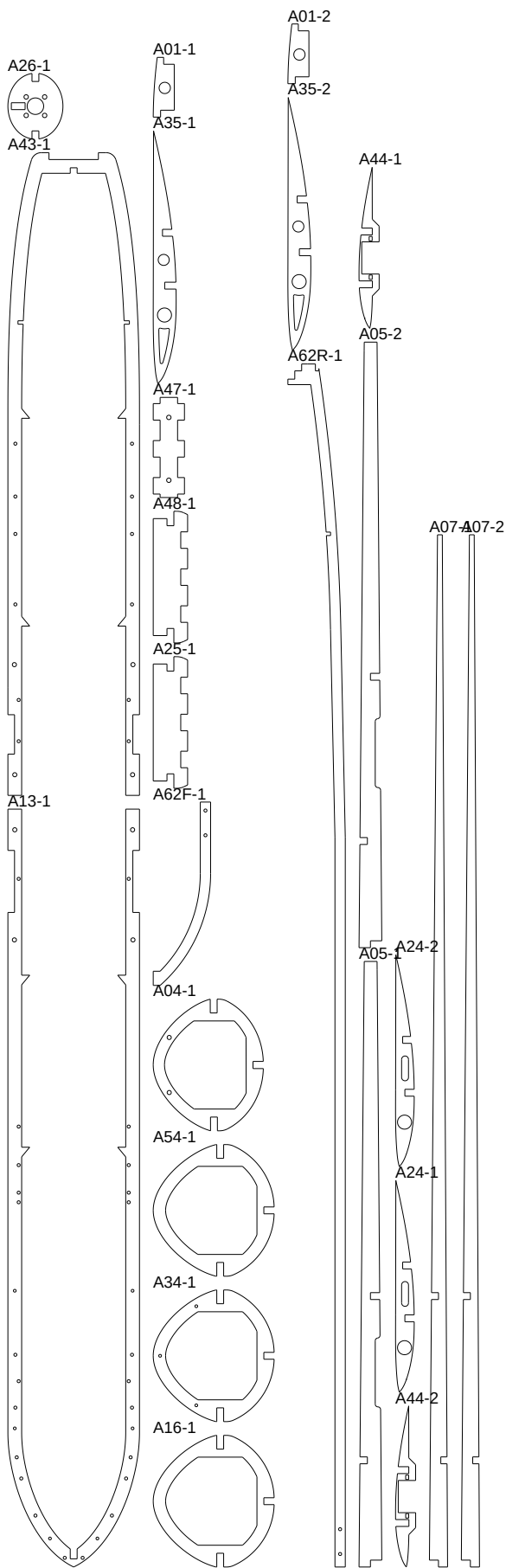


F33 Developed skin

fuselage_src10_face18
5.3 x 151.1 mm | max strain 0.000%

5mm stock: 24 pieces

Reference sheet1220x2440mm. Compact work area 362.8x1141.2mm. Grain +Y.



3mm stock: 6 pieces

Reference sheet1220x2440mm. Compact work area 90.0x703.2mm. Grain +Y.

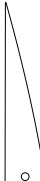
A10-2



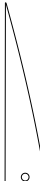
A10-1



A53-2



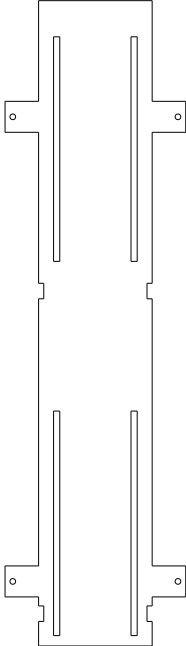
A53-1



G03-1



I01-1



2.5mm stock: 7 pieces

Reference sheet1220x2440mm. Compact work area 101.0x441.6mm. Grain +Y.

A63-2



A63-1



H05B-4



H05B-3



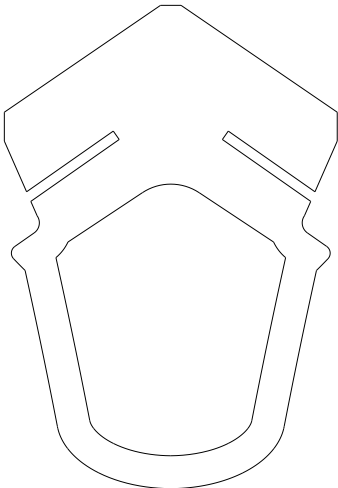
H05B-2



H05B-1

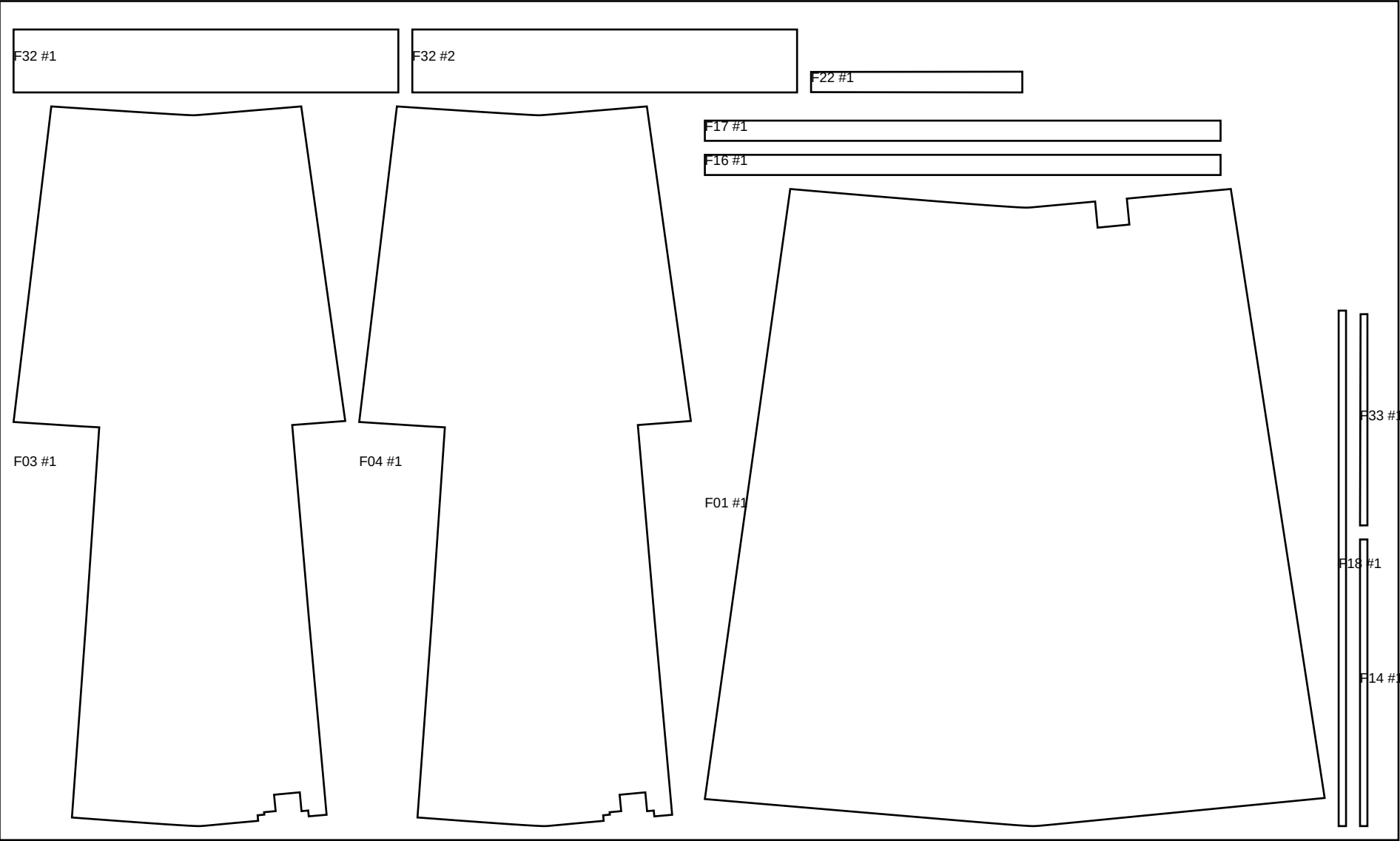


A18-1



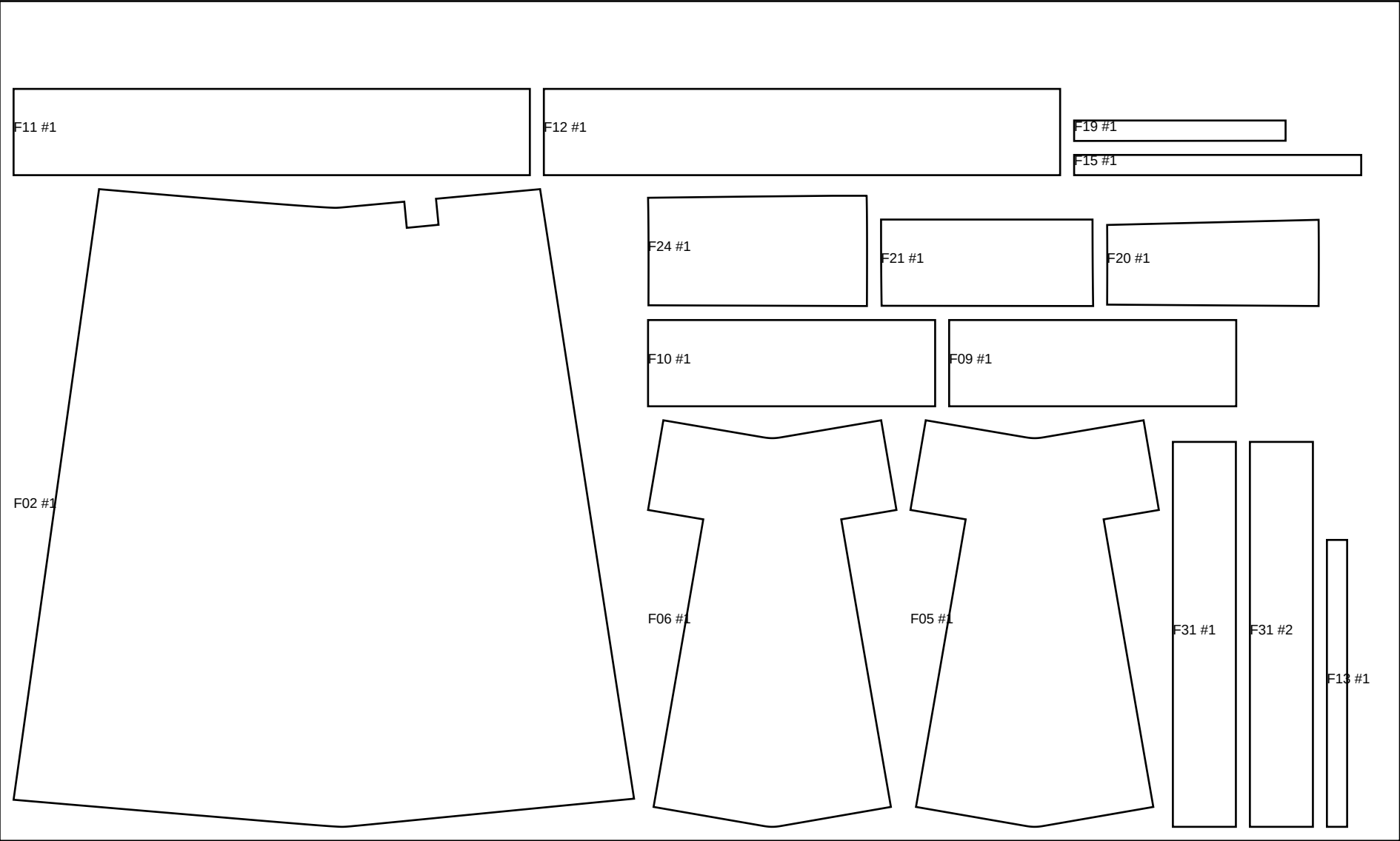
Foam sheet 1 / 1000 x 600 x 5 mm

Overview only. Knife cut at 100% from DXF or full-size pattern tiles. CUT layer only.



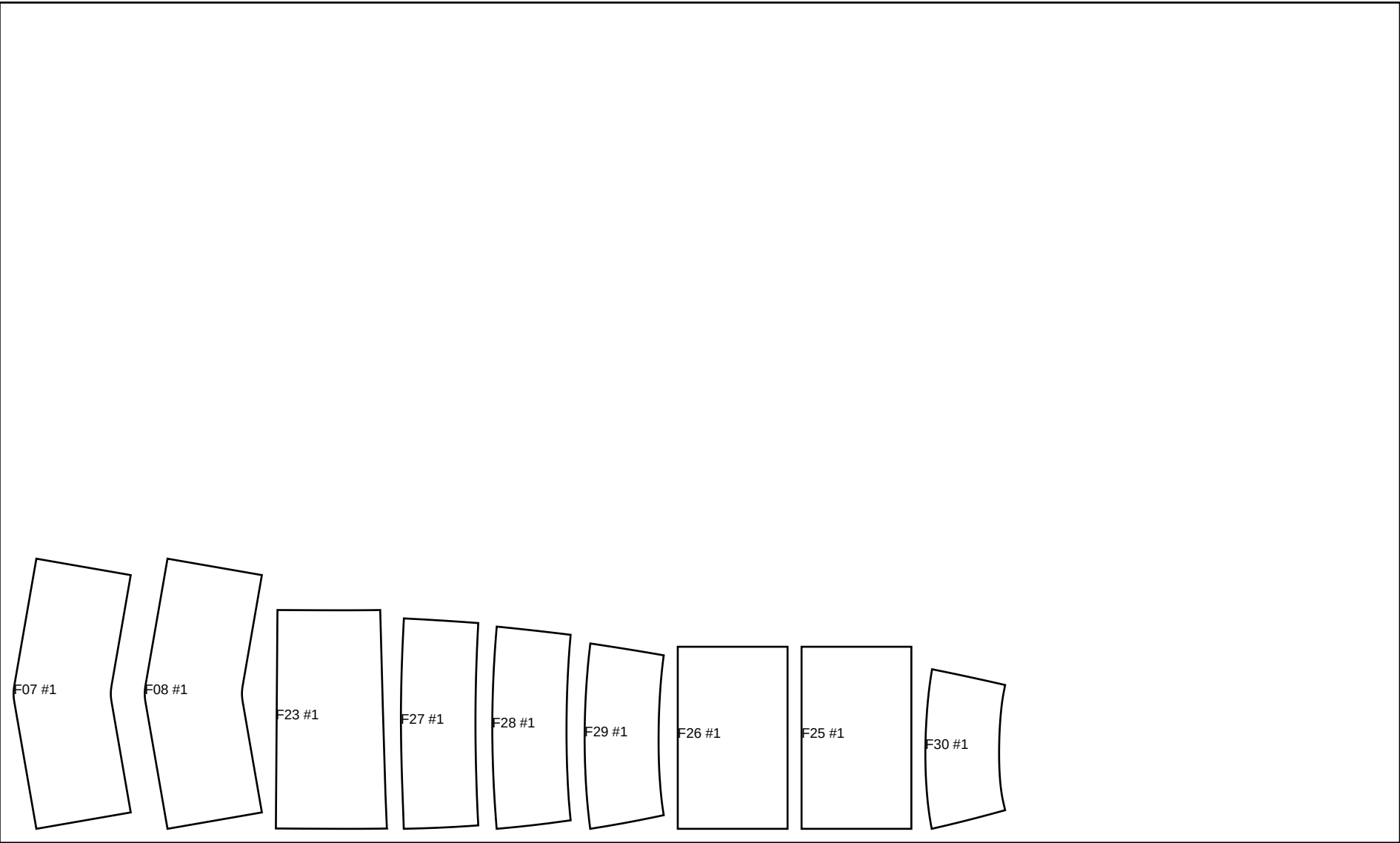
Foam sheet 2 / 1000 x 600 x 5 mm

Overview only. Knife cut at 100% from DXF or full-size pattern tiles. CUT layer only.



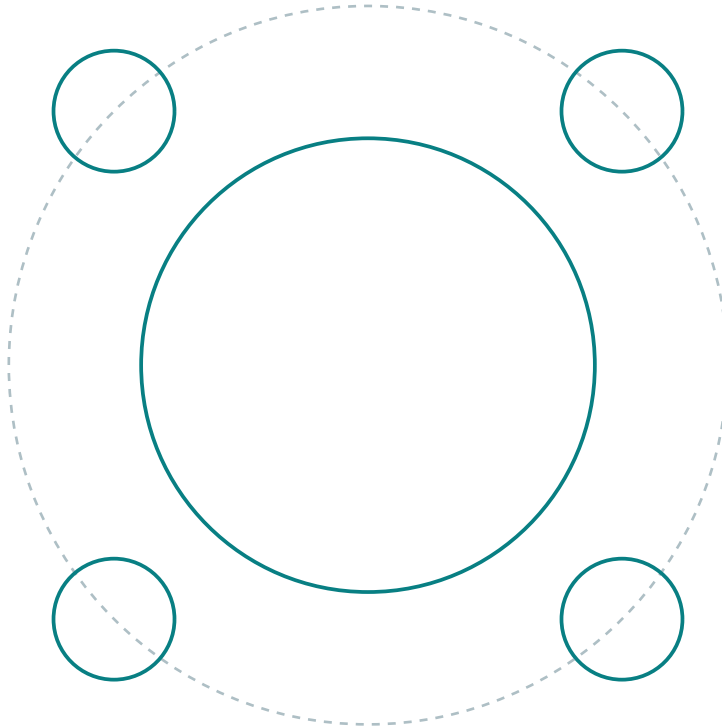
Foam sheet 3 / 1000 x 600 x 5 mm

Overview only. Knife cut at 100% from DXF or full-size pattern tiles. CUT layer only.



A26 / motor mounting

Emax ECO II 2807 1300 KV / 5 mm plywood firewall



Four clearance holes: diameter 3.2 mm on a 19 mm bolt circle

Adjacent hole spacing: 13.435 mm. Centre opening: diameter 12 mm.

Use the A26 DXF or actual-size plywood template for cutting.

This enlarged reference is not a cutting template.

Use four M3 motor screws and washers. Measure safe thread engagement on the actual motor so no screw can reach a winding.

The rear spine ends flush at Y=582.65, the motor mounting face.

No plywood tab may extend behind this plane into the motor envelope.

Keep phase leads secured and clear of the rotating bell.

Checks and build acceptance

File checks passed for the specified revision. Physical results are recorded by the teams during the build.

Digital check	Recorded result
Plywood	25 profiles /37 pieces. Eight changed or added profiles reconstructed; maximum changed-profile volume difference 0.000215%. Original retained profiles keep their recorded edge-finishing requirements.
Printed parts	38 files /42 installed pieces. The 32 new/revised meshes are closed and manifold; maximum accurate CAD-volume difference 0.090%. Six retained aircraft meshes keep their previous checks.
Installed geometry	No detected overlaps for the tested new accessories/gear against the revised static model. The separate structural audit found inherited frame/foam contacts; relieve inside core as instructed. Battery positions, stops and sampled insertion/removal paths checked. Equipment envelopes have separate checks.
Foam	33 developed patterns /35 blanks retained. Prior maximum sampled strain 0.367% and seam mismatch 0.027 mm. New service/gear ports are marked after forming, not projected into replacement flat skins.
A4 plywood	Seven pages: five for 5 mm stock, one for 3 mm, one for 2.5 mm. Whole-page fit, minimum tiling, placement, calibration and paired marks checked.
Reference CG	522.3 mm aft-nose geometric quarter-MAC. Sliced-mass balance estimates span 517.2-532.8 mm aft-nose over 80 mm battery travel. Battery centre Y=-33.7 reaches the geometric reference in this estimate; no flight CG range is approved.

Physical checks still required

Measure stock and coupons. Dry-fit actual hardware, controls and covers. Test forming, joints, print strength and gear deflection. Weigh the completed aircraft and measure CG. The Electronics Team verifies loaded rails, sensors, radio range, failsafes and propulsion current/thrust/temperature.

Limits

Static fit and file export do not qualify structural loads, cooling, flutter, hand-launch performance or flight stability. The shop qualifies its bend process. An experienced fixed-wing pilot reviews the completed aircraft before flight.